



Undergraduate Graduation Thesis



题目: Surgical-YOLO: A Real-Time Intelligent Analysis

Framework for Intraoperative Endoscopic Video Based on Improved
YOLO

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摘要

软件工程

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脊柱微创手术中的术中感知是一项基本挑战：外科医生需在狭窄的内镜视野下实时追踪多个相互重叠的软组织结构，同时满足实时视频帧率的要求。注意力增强的YOLO架构已被证明可提升手术视频分割精度，但现有研究均将单一注意力机制均匀部署至特征金字塔的每一层级，默认各位置受益程度相同。特征金字塔不同层级编码着本质不同的空间与语义信息，是否应针对不同位置采用不同类型的注意力机制，目前尚未得到系统研究。

本文通过对YOLO11n-seg的系统性双阶段实验研究来填补上述空白。YOLO11n-seg是一款实时实例分割架构，其特征金字塔颈部结构具有五个独立的注意力插入位置。探索阶段对六种候选注意力机制在全部五个位置上进行筛查；验证阶段则在包含6,958标注帧的脊柱内镜数据集上，对十种配置以100个训练轮次和10个随机种子进行评估，数据集涵盖六类解剖结构。

核心发现是注意力机制的有效性具有位置依赖性：三元注意力（Triplet Attention）在高分辨率和中分辨率输出层显著提升了性能（L19: +0.96%, $p=0.021$; L23: +1.36%, $p=0.001$ ），但在低分辨率语义层未产生显著效益（L27: +0.11%, $p=0.831$ ）。最优位置感知配置Hybrid-L15CA实现了 $mAP_{50-95}=0.5913$ （相较基线提升+2.82%），在仅修改五个位置中的三个的条件下超越了均匀三元注意力部署，并以191.6 FPS的推理速度运行——比基线快18%。五个解剖结构类别观察到统计显著提升，且效应量优于完全均匀部署。

上述结果表明，位置感知注意力设计是多尺度分割架构中均匀部署的可行替代方案，在提升精度的同时兼顾推理效率，可直接应用于实时手术导航系统。

关键词： 手术视频分割；脊柱内镜；YOLO11；注意力机制；特征金字塔网络；位置感知注意力

ABSTRACT

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Intraoperative perception in minimally invasive spinal surgery remains a fundamental challenge: the surgeon must simultaneously track multiple overlapping soft-tissue structures through a narrow endoscopic field while operating at real-time frame rates. Attention-augmented YOLO architectures have improved surgical video segmentation, but all existing work deploys a single attention mechanism uniformly across every level of the feature pyramid, assuming equal benefit regardless of position. Whether different pyramid levels, which encode qualitatively distinct spatial and semantic information, benefit from different attention types has not been investigated.

This thesis addresses that gap through a systematic two-phase experimental study on YOLO11n-seg, a real-time instance segmentation architecture with five distinct attention positions in its feature pyramid neck. An exploratory phase screens six candidate attention mechanisms across all five positions, and a validation phase evaluates ten configurations at 100 training epochs and 10 random seeds on a spinal endoscopy dataset of 6,958 annotated frames spanning six anatomical classes.

The central finding is that attention effectiveness is position-dependent: Triplet Attention produces statistically significant improvements at the high- and mid-resolution output layers (L19: +0.96%, $p=0.021$; L23: +1.36%, $p=0.001$) but no significant benefit at the low-resolution semantic layer (L27: +0.11%, $p=0.831$). The optimal position-aware configuration within the configurations tested, Hybrid-L15CA, achieves $mAP_{50-95}=0.5913$ (+2.82% over the MSCA baseline), surpasses uniform Triplet deployment while modifying only three of five positions, and runs at 191.6 FPS which is 18% faster than the baseline. Statistically significant improvements are observed across five of six anatomical structures with a larger effect size than full uniform deployment.

These results demonstrate that position-aware attention design is a viable and more efficient alternative to uniform deployment in multi-scale segmentation architectures, with direct applicability to real-time surgical guidance systems.

Keywords: surgical video segmentation, spinal endoscopy, YOLO11, attention mechanism, feature pyramid network, position-aware attention

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Chapter 1 Introduction

1.1 Introduction

The emergence of minimally invasive and percutaneous endoscopic techniques has revolutionized spinal surgery for degenerative disc disease and spinal canal stenosis. Decompression through a self-retaining tube that limits the surgeon's operative corridor to a single working cannula results in significantly less soft-tissue damage, blood loss, and postoperative recovery time than traditional open procedures [1]. However, this comes at the cost of a challenging visual scene. Six different anatomical structures - the intervertebral disc, vertebral skeleton, ligamentum flavum, paraspinal muscle, traversing nerve root, and herniated nucleus pulposus - must be individually recognized and tracked as they overlap in view [2]. Movement of the working instruments through soft tissue dynamically alters their shape. All of this takes place through the narrow lens of an endoscope with limited depth perception, changing illumination conditions due to motion artifacts and pooling blood, and constant occlusion by the instruments themselves. These structures often have similar colors and textures and appear at radically different scales within the same frame. Cognitive load related to the given environment influences operational time as well as intra-operative error rate leading to design impediments to widespread adoption of endoscopic spinal procedures [2].

Real-time computer vision allows for an avenue of reducing cognitive load [3]. An instance segmentation algorithm that can identify any identified structure on a pixel-by-pixel basis throughout a video would allow for continuous position-based feedback to a surgeon, automatic identification of surgical phases for workflow analysis and a perceptual backbone for autonomous surgical guidance. Such a system would need to run at frame-per-second (FPS) rates compatible with live video [3], i.e., typically 25–30fps on operating room compatible hardware [4] and also provide clinically significant segmentation accuracy for small, deformable, visually ambiguous structures. Traditional two-stage detectors can provide high-level of accuracy but are not able to provide a level of speed to maintain a "real-time" performance [5]. Thus, this work attempts to define this fundamental compromise.

This work defines a solution for this issue via three modes of investigation: (1) Clinically, real-time segmentation of tissues that are more accurately segmented may reduce the cognitive load placed on the surgeon while navigating the extremely narrow endoscopic corridor, thus possibly decreasing the duration of the surgical procedure and reducing the incidence of unintentional tissue injury. (2) From an engineering standpoint, any improvements

to accuracy must be made within the specified inference budget of hardware used in the operating room; hence, parameter-efficient attention mechanism will be not only necessary in practice but also desirable for academic research. (3) From an academic standpoint, does the type of attention mechanism vary across the pyramid position, which is an architectural design question that applies to any multi-scale segmentation network that deals with a qualitative difference in feature representation across various levels of hierarchy.

Of the various single stage architectures in YOLO family, YOLOv8 and its successor YOLO11 represent a highly practical approach to the previously stated tradeoff. With YOLOv8 and YOLO11, a well-designed single-stage pipeline outperforms two-stage architectures while achieving similar accuracy but requiring considerably less computational resources [6], [7]. This demonstrates that either architecture could be effectively used in the operating room or other resource constraints. YOLO's notable representational learning ability comes from FPN and PAN neck structures. Both FPN and PAN structures generate analogous three spatial scales of feature map in parallel. The high-resolution features produced at P3 (80×80) contain sufficient spatial detail to allow for accurate delineation of anatomical structures. The mid-resolution features produced at P4 (40×40) achieve the right balance between providing spatial precision for individual anatomical structures and supplying the semantic information needed to maintain anatomical relationships. Finally, the low-resolution features produced at P5 (20×20) contain sufficient semantic information to define the anatomical entity's global context. All three of these feature map scales are supplied to an independent segmentation head, which uses a single forward pass to process the spatially dependent feature maps so that all anatomical structures are segment able. The architecture of YOLO11n-seg, the nano-scale segmentation variant, serves as the baseline architecture for this study, which has been augmented by employing the MSCA at all five attention positions, thus representing the laboratory's-established baseline configuration.

In recent years, the use of attention mechanisms has become a standard approach for improving the representational quality of convolutional architectures without increasing the number of parameters or the time required for inference. Squeeze-and-Excitation networks were the first to employ the principle of recalibrating the channel-wise features via global pooling and learned sigmoid gating [8]. The Convolutional Block Attention Module (CBAM) introduced an extension of the SE network for cross-dimensional feature refinement using the sequential channel and the spatial attention [9]. The Triplet Attention module captures interactions across the three pairs of dimensions (Height $\times$ Width, Channel $\times$ Height, and

Channel $\times$ Width) at once. The architecture of the attention mechanism allows it to capture cross-dimensional information without losing sequential information [10]. This approach to surgical videos has demonstrated improvement over the various unaided baselines collected from laparoscopic, bronchoscopic, and robotic-assisted procedures through the application of an attention-based version of YOLO [11], [12], [13]. As a convention across these architectures, however, all work place the same module across all level of the feature pyramid, with no regard for the potential effects of using different types of attention mechanisms at each of these levels.

The same uniform methodology would be architecturally expedient. However, it is still hard to justify theoretically. The three pyramid levels encode significantly dissimilar information. P3 feature are spatially accurate and can detect boundaries; P4 provides a balance between spatial and semantic information; while P5 primarily conveys semantic information [14], [15]. In contrast, the YOLO11n-seg architecture in this work includes additional levels of attention. Providing five distinct levels of attention within the architecture: L11 (the output of the backbone); L15 and L19 (along the top-down fusion path on the FPN); and L23 and L27 (along the bottom-up aggregation path of the PAN). Each of these levels performs a different function compared to its neighbors. For example, L15 is a fusion layer that absorbs deeper representations from the backbone and combines them with shallower spatial representations before feeding P3 output at L19. The purpose of L15 is to prepare for the L19 rather than predict it. Whether the optimal attention mechanism is itself a function of position has not been systematically investigated in surgical video segmentation or, to the best of the author’s knowledge, in any multi-class medical image segmentation context. The question has two dimensions: whether a fusion layer benefits from different attention than a segmentation output layer, and whether a high-resolution spatial layer benefits from different attention than a low-resolution semantic layer. This gap motivates the central hypothesis of the present work.

This thesis presents a framework, that explores systematic attention deployment based on location within the feature hierarchy of a spinal endoscopic video segmenting network. The core hypothesis is that different locations in the feature pyramid are expected to have different representational attributes and will thus require location-specific mechanisms of attention, rather than uniform attention deployment, as a way to improve performance. To support this hypothesis, we developed a two-phase experimental program for testing. Phase 1 included an initial phase of exploration of 6 non- MSCA candidates (Triplet Attention, Channel Attention (CA), Efficient Channel Attention (ECA), Global Context (GC), SimAM, and Biformer) in the 5 attention locations using 50 training epochs with 3 random seeds to produce the location-

mechanism preference signals needed to create hybrid configurations. After that, in Phase 2 we validated the hybrid configurations using 100 epochs and 10 independent random seeds, in order to have a statistically valid assessment of performance. The assessment of performance was conducted on a spinal endoscopic video dataset containing a total of 6,958 annotated frames, with 6 different anatomical structures. The major finding of this study is that the effectiveness of attention is dependent on its position. For example, Triplet Attention improved both P3 and P4 output layers and did not improve P5, confirming our hypothesis. Hybrid-L15CA configuration achieved an $mAP_{50-95}=0.5913$, which represents a higher score than using the same Triplet deployment with fewer changed modifications in terms of position as well as running 18% faster than the baseline.

1.2 Current Studies or State of the Art

1.2.1 YOLO-Based Surgical Video Segmentation

Originally, YOLO (You Only Look Once) architectures were meant for single-stage object detection, but they have since developed into instance segmentation (IS) with the addition of a mask prediction head, first evaluated in YOLOv8-seg and then in YOLO11-seg [6], [7]. With excellent throughput, competitive accuracy, and simple implementation, YOLO architectures have become the default model for video-based surgical video analysis for many different surgical procedures [3]. For example, LC-YOLOmatch demonstrated the capacity for YOLO-based dense segmentation to achieve automated surgical phase recognition at clinically relevant frame rates during laparoscopic cholecystectomy [16]. Using YOLOv8, the real-time dense segmentation of instruments in endoscopic surgical procedures has opened the possibilities for downstream surgical workflow applications [17]. Similar evaluations of robotic-assisted surgical instruments through YOLO have also demonstrated its unique value for intraoperative instrument recognition [12]. In addition, while the Spine Endoscopic Atlas established a shared benchmark for instrument segmentation in spinal endoscopic surgery [18], the Atlas did not address the multi-class anatomical tissue segmentation issue, which is far denser than the instrument detection. These works collectively validate YOLO's ability to analyze surgical videos, but each has focused on either a particular surgical event (i.e., phase recognition, detection of instruments, and/or segmentation of a surgical scene) or a specific task (i.e., phase recognition) - there is no common multi-class anatomical segmentation framework for spinal endoscopic surgery. While Hybrid CNN-Transformer architectures show relatively superior performance potential [19]; they involve significantly larger associated computational footprints, which prohibit their immediate use in real time within an operating room

environment. It is therefore suggested that a lighter-weight single-stage solution, augmented specifically for the purposes of this investigation, is likely to serve as the optimal solution for surgical video datasets.

When compared to a standard YOLO-based architecture, adding attention modules produces measurable and consistent improvements in performance across several different benchmark categories of surgical video. For example, by employing adaptive multi-scale attention, AMAY-Net achieved substantially improved segmentation of liver and gallbladder boundaries during laparoscopic surgeries [11]. YOLO algorithms that incorporate CBAM as an attention module also demonstrated significantly improved detection accuracy under variable lighting conditions and tissue deformation during pediatric bronchoscopy [13]. By implementing multi-scale hybrid attention designs, significant improvements in detecting complex anatomical structures across medical imaging modalities have also been obtained [20]. All of the above works describe an identical architectural approach in that all employ one selected attention modules placed at each level of the feature pyramid and equally located within the pyramid. None of the studies provide a detailed analysis to ascertain if the performance gains noted were uniformly represented across all levels of the feature pyramid, nor if every position within the pyramid offered an equally level of contribution to achieving the resultant performance gains. There have been no proposals for investigations to the knowledge of this thesis, to ascertain if selective or heterogeneous attention can be employed and can obtain equal or superior results as compared to full uniform employment across the pyramid levels. This means every cited work leaves open the question of whether the performance gains from attention are uniformly distributed across pyramid levels, or whether some positions contribute the bulk of the improvement while others contribute nothing, a distinction that has direct implications for deployment efficiency.

1.2.2 Attention Mechanisms and Feature Pyramid Position

The idea that feature pyramids exhibit representational heterogeneity is not new, and this heterogeneity was the basis for the FPN [14] and PAN [15] designs to take advantage of different levels of information represented in feature pyramids through top-down and bottom-up pathways allowing for explicit features to be combined across levels. For example, the three output layers of P3, P4 and P5 have different representation qualities. Further evidence for the idea of different representational levels playing distinct roles within a single forward pass was presented through the use of bidirectional fusion in later designs [21]. The literature on attention mechanisms has created a number of different attention modules that take advantage of the

representational regimes of the input; channel attention modules such as SE [8], ECA, and MSCA are designed for feature distributions in feature channels, therefore they are naturally suited to semantic layers; spatial attention models such as the spatial branch in CBAM [9] and cross-dimensional Triplet Attention [10] exploit spatial activation patterns, therefore they are best suited for high-resolution boundary detection layers. Furthermore, integrated spatial-channel attention designs such as DA-TransUNet [22] and multi-branch spatial attention architectures [23] combine the advantages of both spatial and channel attention but do so uniformly across the entire network.

Multi-scale attention approaches have been applied at multiple pyramid levels in object detection [24] and remote sensing [25], but these designs vary the spatial scale at which a fixed attention type operates, rather than varying the attention type itself across levels. The possibility that the optimal attention mechanism is a function of pyramid position, that the P3 boundary-detection layer has a different preference from the P5 semantic layer, and that a feature fusion layer like L15 has a different preference from a segmentation output layer like L19 has not been explored in any published work in surgical video segmentation or, to the knowledge of this study, in any multi-class medical image segmentation context. This represents a concrete and testable gap in the current literature. Systematic position-level ablation across multiple candidate attention mechanisms, followed by principled hybrid design based on the observed position-specific preferences, constitutes the methodological novelty of the present work.

1.3 Main Work of This Project

This project proposes and validates a position-aware attention framework for real-time spinal endoscopic video segmentation built on YOLO11n-seg. The central question is whether different positions in the YOLO feature hierarchy benefit from different attention mechanisms, and whether a position-aware hybrid design can match or exceed uniform deployment with greater statistical robustness.

The dataset consists of 6,958 frames of spinal endoscopy videos across 6 anatomical classes (intervertebral disc, spine skeleton, ligamentum flavum, paraspinal muscle, traversing nerve root, and herniated nucleus pulposus). Using YOLO11n-seg, there are 5 positions of attention insertion in its FPN-PAN neck at the backbone end (layer 11), along the FPN fusion path (layers 15 and 19) and along the PAN aggregation path (layers 23 and 27). Each insertion position plays a distinct functional role in the feature hierarchy; thus, the baseline configuration uses MSCA uniformly among all insertion positions.

There were 2 phases in the experimental program. In Phase 1, six non-MSCA candidate mechanisms were tested across all five insertion positions for 50 epochs with three seeds to create preference signals for every insertion position mechanism across 66 training runs. Phase 2 involved selecting and validating 10 combinations over 100 training epochs, using 10 independently random seeds to generate statistically valid performance estimation. Mask mAP50-95 was the primary metric utilized with the addition of paired t-test, Cohen's d effect size, and coefficient of variation to evaluate measurement stability.

The most important findings of this study were that Triplet Attention produced statistically significant increases in performance compared to the other mechanisms evaluated in the P3 (Layer 19; +0.96%, $p = 0.0209$) and P4 (Layer 23; +1.36%, $p=0.0014$) output layers, while minimally impacted and not statistically significantly increased the performance in the P5 (Layer 27; +0.11%, $p=0.8309$) semantic output layer. These findings support the core hypothesis by showing that attention position has a direct effect on the acquisition of features for video segmentation associated with surgery. Of the tested attention positions, Hybrid-L15CA was the optimal position-aware configuration, with an mAP50-95=0.5913 (+2.82% baseline) with only three modifications from baseline.

This thesis is fundamentally based on these three main contributions:

(1) Methodological contribution: A systematic 2 phase investigation of position-dependent deployment of attention mechanisms across all five positions within YOLO11n-seg feature pyramid.

(2) Engineering contribution: A systematic methodology has been established for reproducing an experimental framework based on YAML that integrates C2f wrappers across all configurations, and that allows for deterministic seeding up to level four, and produces statistical output using multiple seeds to allow for reliable differences in the design of sub percent for the various architectural configurations of this dataset.

(3) Empirical contribution: The experimental results obtained using the Hybrid-L15CA configuration have resulted in an mAP50-95=0.5913 (+2.82% above baseline) and demonstrated an 18% faster in inference processing speed versus the MSCA configuration while utilizing only three of the five attentional position sources to create the configuration.

1.4 Organization and Structure

The thesis is organized into six chapters, each of which describes the progress made during each of the project phases.

Chapter 1 (present chapter) presents background clinical context for real-time spinal endoscopic video segmentation, reviews the literature on YOLO-based surgical video analysis and current practices of attention module deployment within feature pyramid networks. The two-phase experimental design and principal findings are summarized.

Chapter 2 provides the technical background and literature review necessary to contextualize the work. It covers the evolution and medical applications of YOLO-based segmentation architectures, the principal attention mechanisms evaluated in this study, and the FPN-PAN feature pyramid structure in YOLO11n-seg. The representational distinction between pyramid levels and the absence of position-specific attention design in the existing literature are formally established as the gap that motivates this work.

Chapter 3 presents the system analysis and proposed design. The YOLO11n-seg architecture and its five attention insertion positions are described in detail. The rationale for selecting Triplet Attention as the primary candidate for position-level ablation is developed from the theoretical properties of each mechanism. The hypothesis that different pyramid positions have distinct attention preferences is formalized, and the Hybrid-L15CA configuration is introduced with a principled account of why each position was assigned its respective mechanism.

Chapter 4 discusses how the experimental framework was created, how the spinal endoscopy dataset was pre-processed and augmented, training hyperparameters for the experiments, the use of a deterministic multi-seed strategy and describes in detail the implementation of Attention Module Types including how each type will be integrated into each of the different configurations of the experiments to ensure that all of the module types are configured identically across all experiment configurations. It discusses the two-phase experimental methodology from the Phase 1 screening to the Phase 2 validation.

Chapter 5 The results of single position ablation experiments confirm the validity of the hypothesis and allow for the establishment of a position-level profile of Triplet Attention responses. The hybrid configuration results test whether positions are additive to or complementary with respect to one another, and the full uniform deployment experiments provide upper bound baselines for comparison purposes. A complete statistical analysis and

robustness comparison of both the Hybrid-L15CA and the uniform baselines are presented for every structure; additionally, a discussion of the implications for clinical application of these findings by structure is provided.

Chapter 6 provides an overview of the contributions of this work, discusses limitations of the current experimental scopes, and provides potential avenues for future research. It also includes the reflection of this project.

1.5 Technical Research Route

The Technical Research Path taken throughout this thesis is illustrated in Figure 1-1 beginning with formulation of the clinical research problem and creation of datasets through to the statistical validation of the optimal solution of the position-aware two part experimental program.

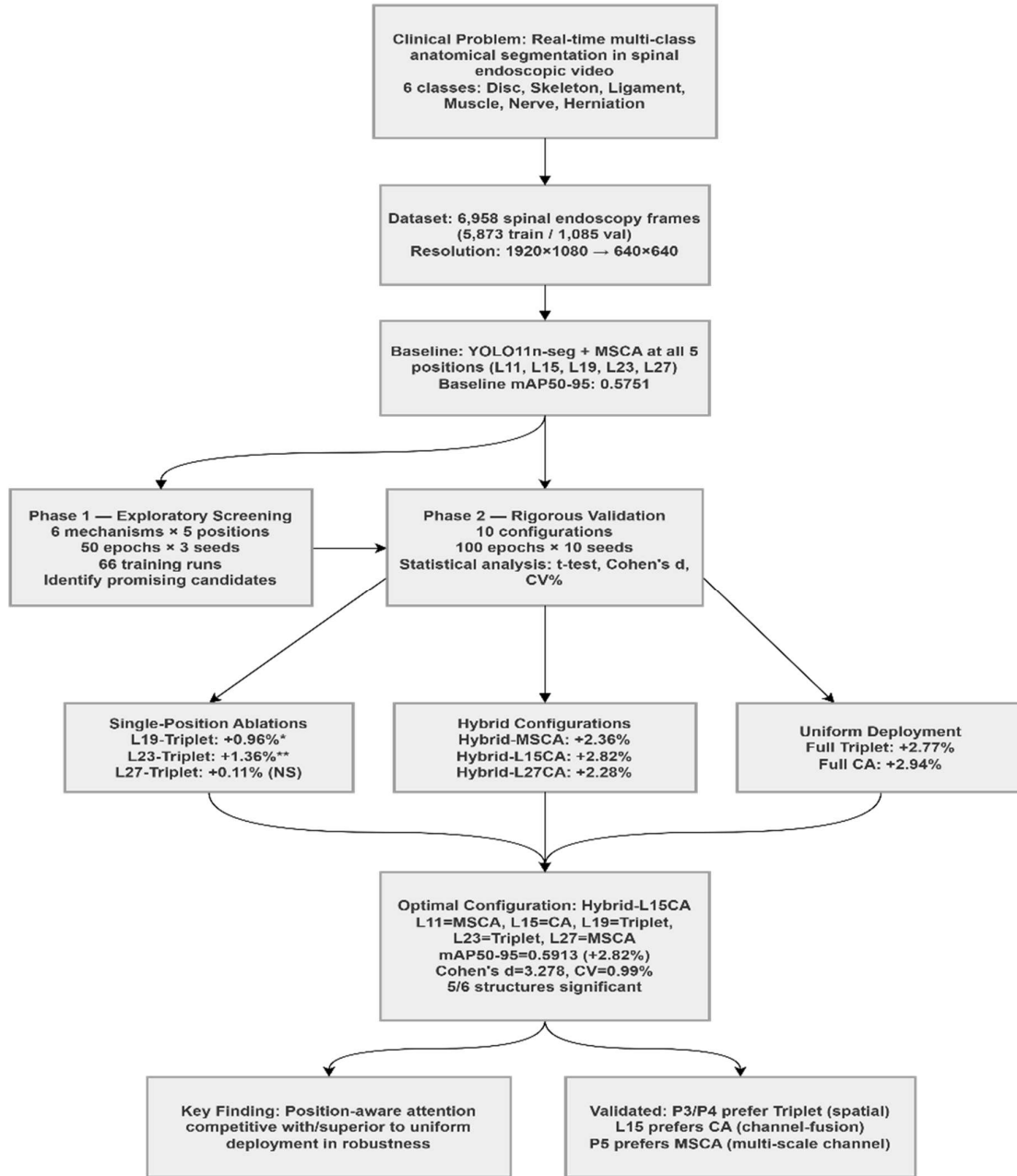


Figure 1-1 Technical Research Route

Chapter 2 Background Knowledge and Literature Review

2.1 YOLO-based Segmentation in Medical Imaging

The YOLO (You Only Look Once) object detection model was originally developed to allow for the detection of objects from images or video in a single pass [26]. Later on, enhancements to the YOLO framework have allowed for the possibility for these detection models to give the ability to perform instance segmentation as well. Through YOLO versions 8 and 11, we can take advantage of nearly real-time inference, along with high-quality accuracy for segmentation purposes, making them suitable foundational components for many real-time medical image analysis systems [6], [7]. Medical applications of YOLO show how effective the YOLO framework is for the segmentation of anatomical structures and the detection of surgical processes over a variety of imaging modalities [27], [28] through both pathology [29], as well as radiology [30]. Advancements in medical image segmentation have been made with the use of hybrid networks that combine CNNs and Transformers within a single model to improve outcomes within the realm of medical image segmentation [31]; as an example, using dual-branch approaches with multi-attention and cross fusion strategies have led to considerably higher performance results through these architectures [19].

Another development is the development of YOLO dense segmentation algorithms to help surgeons analyze surgical video in real time to detect surgical instruments and phases [3]. An example of an automatic detection system for detecting segmental phases of laparoscopic cholecystectomy using the YOLO framework is LC-YOLOmatch [16]. Real time segmentation of endoscopic video using YOLOv8 also allows the use of YOLO for some additional downstream processes (i.e., workflow analysis and guidance) [17]. The YOLO approach has also been demonstrated as being able to detect the surgical tools used in robotic-assisted surgical procedures [12]. Although the Spine Endoscopic Atlas created and defined a list of standard evaluation criteria for many types of surgical tasks, this resource focused solely on instrument detection and provided no methods for solving the problem of segmenting tissues that are overlapping [18].

Increased accuracy of detecting and segmenting using YOLO has occurred as a result of adding attention module to augment the original YOLO detection framework. AMAY-Net used adaptive multi-scale attention to support concurrent segmentation of both the liver and gallbladder while increasing the accuracy of defining the organ edges during laparoscopic surgery [11]. Attention augmented YOLO also provided similar advantages when applied to

bronchoscopic video segmentation [13] , and hybrid attention designs have improved the performance of these models across multiple medical imaging contexts [20]. However, there are no publications to the knowledge of this study that discuss how the overall attentional alignment provided by uniform attention has shifted the performance improvement distributed throughout the pyramid levels of the architecture.

2.2 Attention Mechanisms for Segmentation

Attention modules are designed to place higher importance on informative attributes while diminishing less informative attribute information [32]. Squeeze-and-Excitation Networks pioneered channel attention by global pooling and gating operations [8]. The Convolutional Block Attention Module (CBAM) utilizes channel attention and spatial attention separately through independently, with independent processing for each type of attention. In the CBAM design, features are processed through both attention modules, which together allow the features to benefit collectively from both spatial and channel refinements [9]. Multi-Scale Cross-Axis Attention (MSCA) combines multi-scale receptive field extraction with axial attention, allowing the network to leverage the local and global relationships of features within the channel axis and spatial axes. For the purposes of this work, MSCA will be the foremost type of attention applied in the baseline configuration of YOLO11n-seg. Triplet Attention computes attention on three pairs of the axes through three parallel branches while also capturing cross-dimensional interactions among the spatial and channel axes without losing sequential information as is found in CBAM design [5].

The augmentation of CBAM within YOLO models has aided in segmenting bronchoscopy data [13] but there is a limitation regarding the application of both CBAM's channel and spatial branches across all levels of the pyramid, without considering their representational roles within the pyramid's levels. The use of multi-scale fusion through attention has been proposed as a means of overcoming the challenges presented by the segmentation of complex anatomic structures contained within laparoscopic video [20]. The fusion of spatial and channel-wise attention has proven successful for segmenting multiple types of medical imaging [22]. Additionally, multi-branch architectures employing spatial attention mechanisms have been successful at boundary and edge detection tasks that are critical to accurately delineate anatomical structures and provide enhanced accuracy through feature importance weighting [23]. All of these methods utilized the same attention module across the entirety of the scales for their respective networks, confirming the understanding that feature pyramids produce multiple hierarchical representations of an object in image or video; each

representation provides contextual information of the object. The feature maps produced from high-resolution representations display fine spatial details and object delineation, whereas feature maps from low-resolution representations of higher-level semantics and are better at recognizing global object shapes. Current architectures to the knowledge of this study, treat all pyramid levels identically when incorporating attention, assuming uniform benefit regardless of each level's representational role. However, the representational role of a pyramid position should determine which attention type is deployed there remains unexplored.

2.3 Multi-Scale Feature Pyramids and Position-Awareness Gap

The feature pyramids serve to resolve the problem of scale variance by producing multiple types of pyramid level feature maps, i.e., low-level, mid-level, and high-level, with each type corresponding to a different spatial resolution. FPN uses a top-down approach with lateral connections that allow high-resolution shallow feature maps (low-level) to merge with semantically-rich low-resolution feature maps through up-sampling and element-wise addition [14], [33]. PAN improved on the FPN by allowing data to flow bottom-up through pyramid levels [15], thus improving feature fusion through bidirectional and efficient means [21]. Each of the three types of feature maps have been optimized to analyze objects of various sizes; therefore, small objects and edge detection require the use of higher resolution feature maps (P3 and P4) and large objects will be represented in lower resolution feature maps (P5) containing coarser feature details [34].

The YOLO network combines the FPN and PAN methodologies to provide multi-scale segmentation of objects. The Network produces Feature Maps at 3 pyramid level types: P3 (stride 8), P4 (stride 16), and P5 (stride 32). At a 640 x 640 input size, P3 would provide feature maps of dimension (80 x 80), P4 (40 x 40), and P5 (20 x 20). Each of the 3 levels of pyramid features provides the detection and segmentation heads for the prediction of objects [7]. The backbone of the YOLO network, through a series of down-sampling procedures, extracts features; the FPN layer fuses both semantic information from deep layers and spatial details from shallow layer through a top-down processing approach, while the PAN layer aggregates local features using a bottom-up processing method. Thus, the pyramidal feature maps produced by the YOLO network permitted the segmentation of images having objects of various sizes in one forward pass. Pyramidal Networks enhanced YOLO's performance at the multi-scale, through the use of attention applied at different levels [24], [25], [35]. However, these models maintain the same form of attention across all scales. While feature pyramids have a sense of a hierarchy and features contained in different levels of the feature pyramid are very different

from one another. For instance, features from the third level (P3) provide spatial precision for determining boundaries, features from the fourth level (P4) are a combination of spatial and semantic information to identify mid-level object characteristics, and features from the fifth level (P5) are primarily about semantic information to provide an overall understanding of the features in the scene.

Table 2-1 Comparison of Attention Deployment Strategies in Related Works and the Proposed Approach

Study	Task	Backbone	Attention Type	Pyramid Level Deployment	Limitation
Zhou et al. [11] (AMAY-Net)	Laparoscopic segmentation	YOLOv8	Multi-scale channel	Uniform across all levels	All levels treated identically; no position-level ablation
Yan et al. [13]	Bronchoscopy detection	YOLOv8	CBAM (channel+spatial)	Uniform across all levels	Same module at every pyramid level; no position differentiation
Chen et al. [20] (MedFuseNet)	Medical image segmentation	CNN hybrid	Hybrid channel+spatial	Uniform across all levels	Applied uniformly; position-specific suitability not examined
Sun et al. [22] (DA-TransUNet)	Medical segmentation	Transformer+UNet	Dual-channel+spatial	N/A (encoder-decoder)	Not a multi-scale single-stage architecture; not real-time
This work	Spinal endoscopy segmentation	YOLO11n-seg	Position-specific: CA at L15, Triplet at L19/L23, MSCA at L11/L27	Position-aware (tested per-level)	Single dataset; YOLO11 family only

Note: 'Uniform deployment' means the same attention module is placed at every pyramid level without position-level differentiation. 'Position-aware' means mechanism type varies by pyramid level role. This work is the only entry that systematically varies the attention type as a function of pyramid position.

2.4 YOLO11n-seg Architecture and Attention Positions

Below we describe how architectures based on YOLO and mechanisms of attention are presented in the literature and describe the YOLO11n-seg architecture used for our research by

explaining the architecture in enough detail to support the development of attention position designs for use later in the study.

The YOLO11n-seg architecture is a nano-scale instance segmentation architecture that belongs to the YOLO11 family. There are three main functional components to this architecture: the first is a convolutional backbone (used for hierarchical feature extraction); the second is the FPN and PAN neck (used for feature fusion across multiple scales); the third is the head (used to generate instance segmentation output by combining prototype segmentation masks with predicted coefficients). The backbone progressively downsamples input dimensions and increases channel depths through a series of Conv and C3k2 blocks. In Table 2-2, we present the channel dimensions and spatial resolutions of the backbone key architecture components used in the YOLO11n-seg.

Table 2-2 Backbone feature map output channel dimensions and spatial resolutions at each pyramid stage in the YOLO11n-seg configuration used in this work

Stage	Key Layers	Channels	Spatial Resolution
P1/2	L0: Conv	64	320×320
P2/4	L1: Conv, L2: C3k2	256	160×160
P3/8	L3: Conv, L4: C3k2	512	80×80
P4/16	L5: Conv, L6: C3k2	512	40×40
P5/32	L7: Conv, L8: C3k2	1024	20×20
Terminus	L9: SPPF, L10: C2PSA	1024	20×20

Note: Channel values throughout Section 2.4, reflect the base architecture specification as defined in the model configuration file. For the nano variant used in this work, all channel dimensions are scaled by a width multiplier of 0.25 at runtime.

The FPN top-down path begins at the backbone terminus. L11 is the final backbone output before neck processing begins and serves as the first attention position. L11's (1024-channel, 20×20) feature map is upsampled and concatenated with L6 (512 channels, 40×40), producing a 1536-channel tensor that a subsequent C3k2 block reduces to 512 channels at 40×40. The second attention position L15 is inserted at this point, before the next upsample. L15 features are then up sampled and merged with L4 (512 channels, 80×80), forming a 1024-

channel tensor that a C3k2 block reduces to 256 channels at 80×80 . The third attention position L19 is placed here. This is the P3 output and a direct input to the segmentation head. Table 2-3 contains a summary of the five different attention mechanism's position.

Table 2-3 The five attention positions in YOLO11n-seg and their functional roles

Position	Location	Channels	Resolution	Functional Role
L11	Backbone terminus	1024	20×20	Backbone consolidation after C2PSA; before neck fusion begins
L15	FPN fusion layer	512	40×40	Feature preparation after P4 fusion; feeds into P3 output L19
L19	FPN output / P3	256	80×80	High-resolution output; direct input to Segment head
L23	PAN output / P4	512	40×40	Mid-resolution output; direct input to Segment head
L27	PAN output / P5	1024	20×20	Low-resolution semantic output; direct input to Segment head

The PAN bottom-up path reuses intermediate neck features. L19 is downsampled by convolution to 256 channels at 40×40 and concatenated with L14 (512 channels, 40×40), forming a 768-channel tensor that a C3k2 block consolidates to 512 channels. The fourth attention position L23 sits at this 40×40 output, the P4 scale and a second direct segmentation head input. L23 is then downsampled by convolution to 512 channels at 20×20 and concatenated with L11 (1024 channels, 20×20), producing a 1536-channel tensor. The last of the C3k2 blocks generates 1024 channels at a resolution of 20×20 , which is where the L27 head positioning is located — the output of P5 and the third direct input into the segmentation head.

Three of the five positions (L19, L23, and L27) are direct inputs to the segmentation head and carry the final representational content used for prediction at each pyramid scale. The L11 position is the last position before the start of the "neck" function of the feature pyramid. It serves as a 'bottle-neck' to consolidate features from the different levels of the base layer rather than a prediction-aligned output. L15 sits on the top-down path of FPN after the fused features of P4 have been merged but before those features have been upsampled to produce P3 that are output at L19. Therefore, L15 is the only one of the five positions that directly conditions the input to a downstream layer L19's attention output, thus creating an opportunity for potential complementarity between mechanisms placed at L15 and L19.

2.5 Attention Mechanisms: Technical Foundations

The previous sections introduced attention mechanisms from a literature perspective. This section describes the computational principles of the four mechanisms carried forward to Phase 2. The mechanisms selected for Phase 2 validation were as follows: MSCA (baseline), Triplet Attention, Coordinate Attention (CA), and ECA. The mechanisms chosen to represent the various forms of global context aggregation, parameter-free attention, and sparse routing were GC (Global Context), SimAM (Simple Attention Module), and Biformer (Bi-level Routing Attention). However, none of these mechanisms displayed consistent improvement signals at the 50-epoch stage through the structure of attention positioning. All four mechanisms are included through a uniform C2f-style wrapper. The bypass for the attention module replaces the first processing block inside the wrapper, while both entry convolutions and exit convolutions remain the same across all configurations so that comparisons between mechanisms can be made fairly.

2.5.1 Multi-Scale Cross-Axis Attention (MSCA)

The MSCA is located in all five locations in the supervisor-provided baseline configuration. As the reference for all comparative testing on experimental runs, the MSCA will be used in both phases of this study. It operates in two stages. The first stage includes a multi-scale spatial frontend, which includes three separate branches, each of which utilizes pointwise followed by depthwise convolution at various kernel sizes (3x3, 5x5, and 7x7), in parallel to derive features from progressively larger receptive fields. The outputs of each of these branches gated by a learnable scalar weight that passes through Sigmoid Activation for bounding their contributions from 0 to 1. The gate scalar weight for each branch then concatenated and combined into a single multi-scale feature map using 1x1 convolution. The

second stage utilizes axial attention along the height and width axes of the combined multi-scale map that resulted from the previous stage. The Query, Key, and Value Tensors will be established from a shared linear projection, and the computation of scaled dot-product attention will occur separately along both axes. Finally, the combination of the two axis outputs will occur by utilizing the per-channel fusion learnable weights normalized via softmax. The outcome of this process will be added back to the original input through a residual connection.

$$F_{out} = x + BN(\alpha_h \cdot Attn_h(F_{ms}) + \alpha_w \cdot Attn_w(F_{ms})) \quad (2.1)$$

where F_{ms} is the multi-scale fused feature map, $Attn_h$ and $Attn_w$ are scaled dot-product attention operations along the height and width axes respectively, α_h and α_w are per-channel learnable fusion weights with softmax normalization across the two axes, and BN denotes batch normalization.

2.5.2 Triplet Attention

Triplet Attention captures cross-dimensional interactions across all three axis pair combinations, $H \times W$, $C \times H$, and $C \times W$, through three parallel branches [10], [36]. The three branches of Triplet Attention work independently and on different views of the input tensor by rotating them into the $H \times W$ plane and bringing the target dimension pair. The channel dimension is compressed Z-pooling by taking both the maximum & mean of each channel and concatenating them into a two-channel representation map. A 7×7 convolution followed by sigmoid activation generates a spatial gate, which is applied element-wise to the branch input. The three branch outputs are averaged to produce the final representation. The learnable parameters of the module are confined to the 7×7 convolutional gates in each branch, plus their associated batch normalization statistics, making Triplet Attention parameter-efficient relative to its cross-dimensional representational scope. In contrast to sequential attention designs such as CBAM, which process channel and spatial attention in separate passes, Triplet Attention models all three dimensional interactions simultaneously.

$$F_{out} = \frac{1}{3}(A_{HW}(x) + A_{CH}(x) + A_{CW}(x)) \quad (2.2)$$

where A_{HW} , A_{CH} , and A_{CW} denote the attentive outputs of the $H \times W$, $C \times H$, and $C \times W$ branches respectively, each computed as $x_{bran} \odot \sigma(Conv(ZPool(x_{branc})))$, and σ denotes the sigmoid function.

2.5.3 Coordinate Attention (CA)

Coordinate Attention decomposes spatial attention into two one-dimensional operations along the height and width axes [37]. Two parallel adaptive average pooling operations encode the input along each axis separately. Average pooling along width results in a strip of shape (B, C, H, 1) that encodes height-directional statistics. Average pooling along height results in a strip of shape (B, C, 1, W) containing statistics of features along width. The width strip is transposed and both strips are concatenated along the spatial dimension to form a combined encoding of shape (B, C, H+W, 1). A shared 1×1 bottleneck, followed by batch normalization and SiLU, compresses this joint encoding. The bottleneck output is then split back into height and width components, and two separate 1×1 convolutions generate sigmoid-gated attention maps a_h of shape (B, C, H, 1) and a_w of shape (B, C, 1, W). The final output is the element-wise product of the original input with both attention maps.

$$F_{out} = x \odot \sigma(f_h) \odot \sigma(f_w) \quad (2.3)$$

where f_h and f_w are the height- and width-directional attention logits derived from the shared bottleneck, σ denotes the sigmoid function, and $\odot$ denotes element-wise multiplication applied in a broadcast fashion to the (B, C, H, W) input tensor.

2.5.4 Efficient Channel Attention (ECA)

ECA performs channel recalibration without the dimensionality reduction bottleneck of SE-Net [38]. Global average pooling aggregates spatial information to a channel descriptor of length C. A 1D convolution with an adaptively determined odd kernel size k is then applied to this descriptor, capturing local inter-channel interactions across a neighborhood of k adjacent channels. The kernel size is determined by two hyperparameters γ and b via $k = t$ if t is odd, else $t+1$, where $t = \lfloor (\log_2(C) + b) / \gamma \rfloor$. With the defaults $\gamma=2$ and $b=1$, this ensures that k scales modestly with C . The 1D convolution output is passed through sigmoid and the resulting weights are applied element-wise to the input feature map. Because ECA operates purely on channel statistics with no spatial component, its representational inductive bias differs fundamentally from the spatial and cross-dimensional mechanisms above. In this work, this purely channel-based design motivated evaluating ECA specifically at low-resolution semantic positions, as a hypothesis to be tested experimentally.

2.6 Evaluation Metrics and Statistical Validation Framework

2.6.1 Segmentation Performance Metrics

Model performance is evaluated using mean Average Precision (mAP), the standard metric for multi-class detection and instance segmentation benchmarks [39]. Two mAP variants are reported throughout this work: mAP50 and mAP50-95. Because stricter IoU thresholds penalize both imprecise boundary delineation and spatial localization errors, mAP50-95 is the more demanding metric and is the primary measure reported throughout this work. mAP50 is reported as a secondary diagnostic metric to distinguish whether performance differences between configurations arise at the detection level or at finer localization.

The YOLO11n-seg head produces separate bounding-box and mask prediction branches, so both bounding-box AP (B) and mask AP (M) are computed independently. Precision and Recall are tracked at the epoch level to monitor training convergence.

2.6.2 Statistical Validation Framework

Any individual training run carries noise from random weight initialization, data order, and hardware scheduling, making it an unreliable estimate of a configuration's true performance. Phase 2 therefore consists of training each configuration across 10 independent random seeds, and reporting mean $\pm$ Standard Deviation of the mAP50-95 as the main estimator of the performance. The Coefficient of Variation (CV) of the mAP50-95 will be calculated by taking the ratio of the Standard Deviation to the mean, expressed as a percentage. The CV will serve as a dimensionless evaluation of training stability that can be compared across configurations that exhibit varying absolute performance levels, with all configurations having a CV at or below 1.5% considered sufficiently stable for the purposes of this study.

$$CV = \left(\frac{\sigma}{\mu} \right) \times 100\% \quad (2.4)$$

where σ is the standard deviation and μ is the mean mAP50-95 across 10 seeds.

Using a paired t-test, we can determine if the improvements seen when using the candidate configuration over the baseline are statistically significant [39]. The paired t-test compares the distributions of mAP50-95 from each of the ten independent seeds for both the baseline and the candidate configuration. The reported P-value is the two-tailed probability using the null hypothesis of equal means, calculated using `scipy.stats.ttest_rel` for paired samples. This work has adopted $P < 0.05$ as the threshold value for statistical significance. We

acknowledge that the assumption of approximate normality is standard practice at this sample size in deep learning Ablation Studies [39] but does not constitute a formally valid claim of the distribution of sample values. Each independent seed produces a statistically independent run; therefore, we have met the assumptions of independence for this study. Cohen's d indicates the magnitude of the effect size by the difference in means divided by the pooled standard deviation [40], providing a dimensionless measure of practical significance that cannot be conveyed through P-value alone. A configuration may be $P < 0.05$ significant; however, the Cohen's d value may indicate that the improvement is minimal when compared to the naturally occurring variance between the runs. A large Cohen's d value signifies that the performance gap is significant and reproducible.

$$d = \frac{(\mu_1 - \mu_2)}{s_p} \quad (2.5)$$

where s_p is the pooled standard deviation of the two 10-seed distributions. By convention, $d \geq 0.8$ is considered a large effect.

This three-tiered framework allows us to establish the degree of performance, stability of training and statistical significance of effects. Simply reporting the best result from a single run or selecting the best Seed after the fact results in an optimistic estimate of a configuration's true and reproducible performance; thus, using a 10-seed protocol results in ranking reported performance of configurations independent of the Method of Initialization.

2.7 Research Objectives

The proposed research objective is to find out whether or not different types of individual attention mechanisms can provide greater benefits when used in relation to a feature pyramid position than a single mechanism when implemented uniformly throughout all position. This need for investigation leads to the following set of questions regarding the use of position-aware attention methods in the area of segmentation of surgical videos:

1. What is the relationship between the different types of attention mechanisms used at different locations within the feature pyramid?
2. How does implementing hybrid attention perform compared with the uniform implementation across all levels of the pyramid?
3. How does the use of position-aware attention methods provide advantages when used in the area of video segmentation of endoscopic spinal video?

4. What benefits can the use of combinations of validated position-aware attention mechanisms provide?

To address these questions, an organized set of experimental systems was set up. The experiment will be carried out in two phases. In the first phase, all of the combinations possible between six potential attentions across the five positions within the feature pyramid will be tested across 50 epochs of training and 3 independent seed runs, to identify the best combinations of attention mechanism and positional choice. In the second phase, an analysis of the top combinations found in phase one will be carried out through training for 100 epochs using 10 independent randomly chosen seed runs using the results of the MSCA as a baseline for each position. Furthermore, based on spinal endoscopic video frames used as the origin of the dataset consisting of six anatomical structures, each type of video frame will be examined using the anatomical structures and also through the aggregation of all results.

2.8 Chapter Summary

Chapter 2 contains the foundational knowledge, technical knowledge, and logical development needed to establish a basis for the experimental designs of this work. Sections 2.1 to 2.3 present an overview of how surgical videos are segmented using a YOLO-based architecture, the development of the YOLO architecture and the investigation of how various attention mechanisms have been implemented in other work. Sections 2.4 and 2.5 provide an overview of the YOLO11n-seg architecture and corresponding channel dimensions and spatial resolutions for every layer of the YOLO11n-seg architecture, in addition to how attention mechanisms are distributed throughout the feature pyramid of the YOLO11n-seg architecture. In Section 2.6, the various evaluation metrics and the statistical validity testing methods utilized are discussed. The application of the systematic approach discussed in Chapter 2 will be presented in the following chapter.

Chapter 3 System Analysis and Design

3.1 System Requirements Analysis

3.1.1 Overview of System Requirements

The main engineering goal of this project is to combine a real-time ongoing spinal endoscopic video feed with the creation of a system that can identify and outline six anatomical regions (Intervertebral Disc, Vertebral Skeleton, Ligamentum Flavum, Paraspinal Muscle, Traversing Nerve Roots, and Herniated Nucleus Pulposus). This project aims to develop a system that provides sufficient segmentation accuracy for intraoperative guidance during surgeries, using standard operating room equipment. The design of the system must be repeatable for the evaluation framework of the five points where attention is inserted when creating the feature hierarchy of YOLO11n-seg.

There are three main requirements that make up the requirements for this system. The first requirement pertains to segmentation; therefore, the system must identify and segment six different classes, each of which will vary significantly in terms of structural appearance due to their differing visual scales, textures, and occurrence frequency. Due to the endoscopic imaging conditions, there will be no depth perception; in addition to this, changing illumination levels and instruments blocking the view will pose additional problems for segmentation. The second type of requirement is the requirement for deploying in real-time; therefore, the inferences for live surgical video will need to occur at the frame rate using standard operating room equipment. Finally, the third type of requirement is experimental validity; for the segmentation performance for the various training configurations to be compared objectively and independently of stochastic initialization variations, all training configurations must be trained under the same conditions.

The design of Surgical-YOLO provides a solution to all three requirements by leveraging three design decisions: to utilize the YOLO11n-seg architecture, which has demonstrated an ability to perform real-time single-stage segmentation; to add attention to five different structural locations of the feature pyramid neck by enclosing each attention module in a common uniform C2f wrapper to ensure all objects tested are comparable; and to implement a deterministic multi-seed training scheme to obtain statistically valid performance estimates. The development of the Surgical-YOLO system creates an actual implementation in this chapter of the research hypotheses proposed in Chapter 1.

3.1.2 Analysis of System Data

The research data set utilized for this study is created from raw, high-definition spinal endoscopic video captured from clinical practices. Video frames were extracted at a resolution of 1920×1080 pixels from surgical videos, and then annotated by Clinical Experts via an instance segmentation mask at the polygon level. The raw annotation files were stored in LabelMe JSON format with each annotation file containing one or more polygon-shape entries that contained Chinese anatomical labels which needed to be normalized and mapped to standard English class identifiers.

A custom Python script (segmentData.py) provides a pipeline for the preparation of this data. We parsed each of the JSON annotation files and then processed them using a label normalization process to resolve orthographic variants in Chinese, including using full and half-width parentheses, inconsistent spacing, and synonym labels, mapping them to 6 canonical categories via a 2-level lookup table. Next, we validated the polygon geometry (minimum of 3 vertices and all coordinates must be located within the image bounds), converted the polygon coordinates to the normalized YOLO segmentation format, and assigned each frame to either the training or validation set by applying a deterministic filename hash based on a ratio of approximately 85:15. After normalization, frames with no valid annotations remained as hard-negative training samples with blank labels.

After preprocessing, the dataset contained approximately 6,958 annotated frames, with 5,873 allocated for training and 1,085 allocated for validation. Table 3-1 contains the number of annotated frames in each class for each of the splits, indicating a significant imbalance among the classes that will affect the design of the training process and interpretation of results.

Table 3-1 Per-Class Annotation Instance Counts in the Spinal Endoscopy Dataset

Class	Train Count	Val Count	Total	Train Prop.	Val Prop.	Class ID
Disc (Intervertebral Disc)	1,294	240	1,534	14.6%	14.6%	0
Skeleton (Vertebral Skeleton)	949	154	1,103	10.7%	9.4%	1
Ligament (Ligamentum Flavum)	3,001	552	3,553	33.9%	33.6%	2
Muscle (Paraspinal Muscle)	202	47	249	2.4%	2.9%	3
Nerve (Traversing Nerve Root)	2,964	556	3,520	33.5%	33.9%	4
Herniation (Herniated Nucleus Pulposus)	434	92	526	4.9%	5.6%	5
Total	8,844	1,641	10,485	100%	100%	—

Note: Train Prop. and Val Prop. are proportions within each respective split.

The distribution of class instances illustrates that there is a large disparity between classes that falls in three tiers. Ligament and Nerve represent the two most annotated classes and make up 67% of all instances in the training dataset. The classes Disc and Skeleton make up a small portion of approximately 25% of all instances combined. There is a very low representation between the classes Muscle and Herniation with 2.4% and 4.9%, respectively, of cases in the training dataset; Muscle has less than 250 instances documented throughout the entirety of the dataset. This imbalance is important because it demonstrates the need for statistical analysis in the per-structure used in Chapter 5; if only aggregate mAP50 to 95 are reported, clinicians would be unable to appreciate how significant the improvements in the performance of minority classes [41], such as Herniation, are when making surgical decisions, despite how critical it is to accurately identify a Herniation during surgery. The proportions of the classes remain consistent across both the training and validation dataset as shown in Table 3-1; this is due to the deterministic filename-hash assignment rather than explicit steps that ensure stratification.

Figure 3-1 shows a sample of annotated frames from the training dataset. The annotated frames show that the segmentation task is visually complex; the view around the endoscope is circular, with a black border that was excluded from annotations. Many of the structures overlap; the disc and nerve root frequently occur together in the same frames. Many of the

structures within each class also have visual differences due to the variation in light angles, the surgical phase, and the condition of the tissues. Ligaments and discs share a similar pale fibrous texture, while nerve roots have less visibility and may be obscured from view by surgical tools or adjacent structures. These factors collectively justify the attention-based architecture design described in subsequent sections.

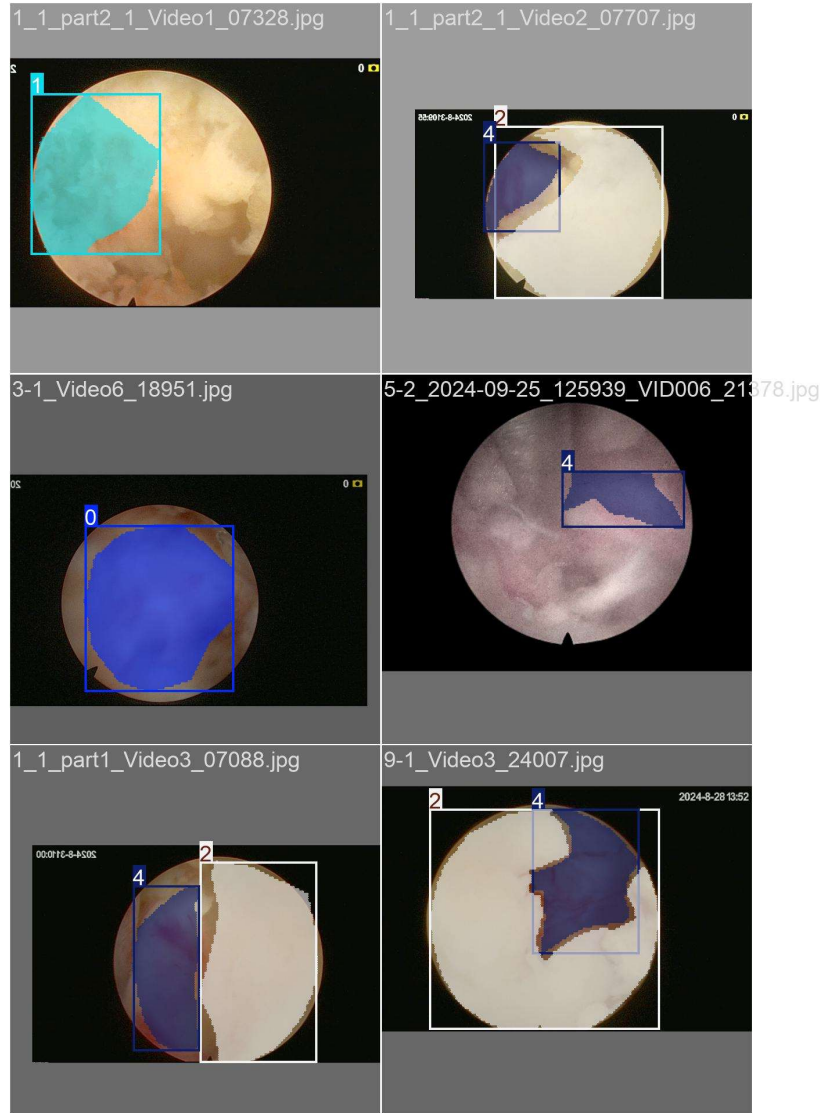


Figure 3-1 Representative Training Batch Showing Annotated Spinal Endoscopy Frames (Class IDs: 0=Disc, 1=Skeleton, 2=Ligament, 3=Muscle, 4=Nerve, 5=Herniation)

Figure 3-2 shows ground-truth segmentation masks for a representative validation batch, with polygon instances rendered in distinct colors alongside bounding boxes. The repeated co-

occurrence of disc and nerve root in consecutive frames from the same surgical sequence is a consequence of the video source's temporal continuity. This is also the reason filename-based deterministic splitting was preferred over random frame sampling. Random assignment would risk placing frames from the same procedure in both splits, introducing data leakage; the hash-based approach prevents this by keeping any single sequence on one side of the split.

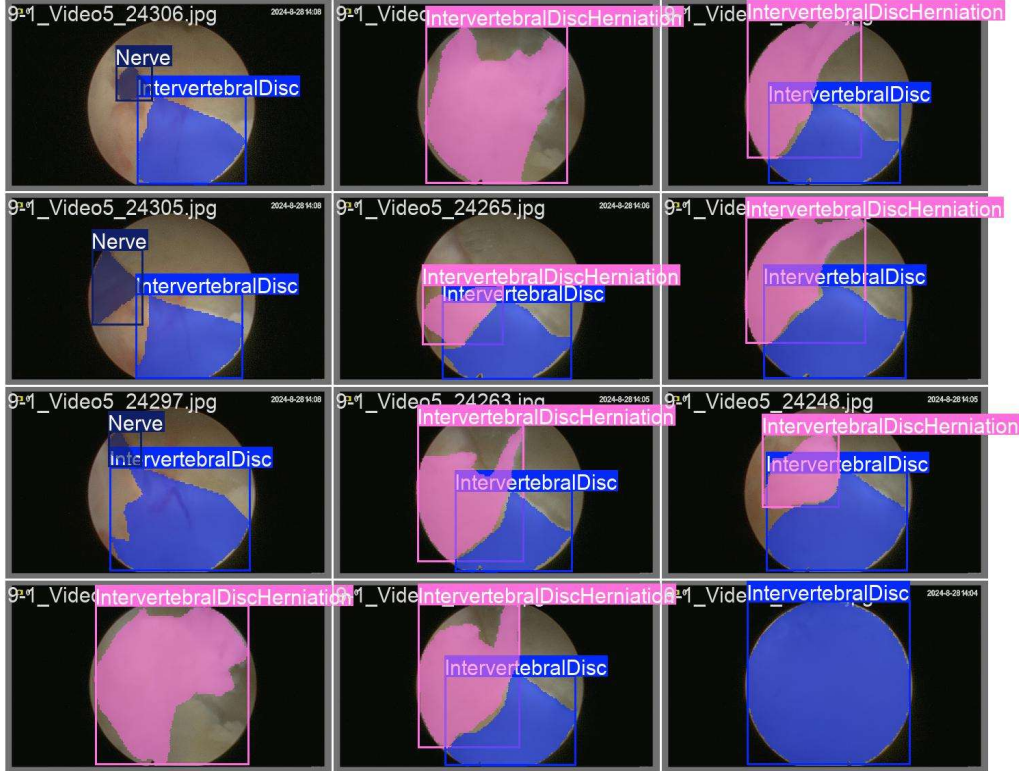


Figure 3-2 Ground-Truth Instance Segmentation Masks on Representative Validation Batch

The data pipeline outputs a standard YOLO directory structure with images/ and labels/ subdirectories per split and a data.yaml specifying class count and names. Images are resized from 1920×1080 to 640×640 during training through the Ultralytics preprocessing pipeline, which pads with letterboxing to preserve aspect ratio.

3.1.3 Analysis of System Function

The functional scope of Surgical-YOLO encompasses four distinct operational modes that together constitute the complete experimental and deployment lifecycle of the system.

The primary function is multi-class instance segmentation inference. Given a single RGB frame at 640×640 resolution, the system produces for each detected anatomical structure

a bounding box with confidence score and a binary instance segmentation mask at effective 160×160 resolution. The segmentation heads work together to detect structures that cover wide size range simultaneously across three pyramid scales: P3 (80x80), P4 (40x40), and P5 (20x20). For each scale, we produce up to 300 prediction/images per scale, all containing information regarding class identity, confidence, bounding box coordinates, and mask coefficient vector (i.e., mask constructed from a pre-trained prototype basis consisting of 32 components).

The second functionality is the multi-seed training pipeline. This allows training a model multiple times defined in a YAML-based architecture file that lists the order of layers along with which attention modules should be associated with those layers. It allows for 50 epochs of Stochastic Gradient Descent (SGD) in Phase 1 of filtering and 100 epochs of SGD in Phase 2 of filtering, utilizing the YOLO framework developed by Ultralytics; and records information about the mean Average Precision (mAP) at both 50% and 95% overlaps for every epoch, which is written to a CSV file for results. The multi-seed training function employs deterministic seeds in each PyTorch, CUDA, NumPy and Python random module prior to the start of model training, allowing for performance comparisons of models trained with multiple different seeds to reflect the inherent random nature of training. This function is executed three times for each configuration in Phase 1 of filtering, and 10 times for each configuration in Phase 2 of filtering.

The third functionality is statistical performance evaluation, whereby the mean and standard deviation values of the 10 mAP50-95 scores for a given configuration (per seed) can be combined to compute the overall mean and standard deviation, as well as the coefficient of variation and t-test statistics (pairwise) against the baseline, using the information stored in the CSV logs generated during training. The statistical comparison tables included in Chapter 5 depict this evaluation.

The fourth functionality is data preparation. The segmentData.py pipeline converts raw LabelMe JSON Annotation Files and the corresponding JPEG Images into the YOLO-format dataset outlined in section 3.1.2. The fourth function (Data Preparation) represents a one-time preprocessing procedure that is not an element in the inference system, so, therefore, this functional decomposition ensures that inference, training, evaluation, and data preparation are separate entities, which aids software engineers in validating and replacing any component independently of all experimental variables.

Figure 3-3 depicts the Functional workflow of the overall system. It traces the raw surgical video from raw annotation through dataset preparation, multi-seed training, and the final statistical evaluation.

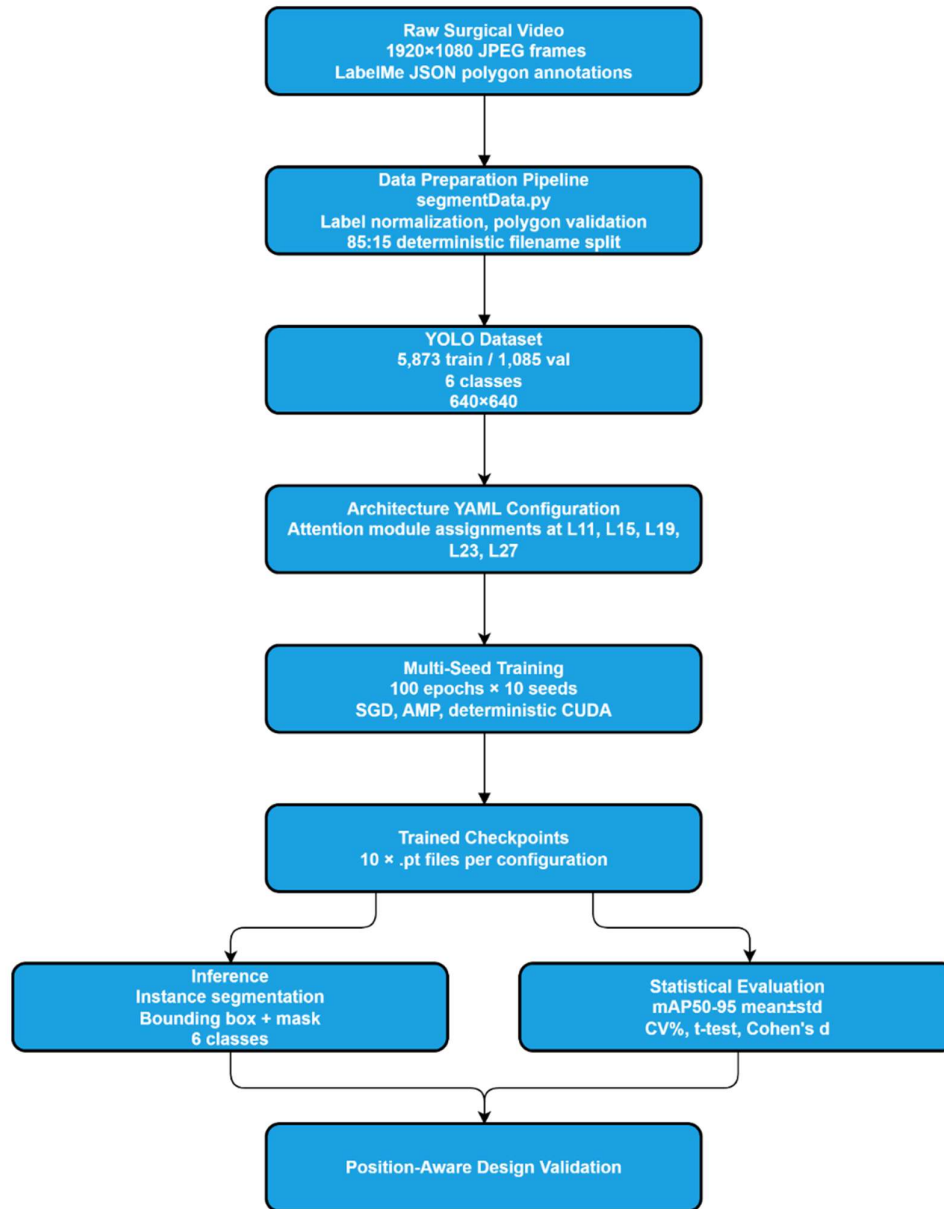


Figure 3-3 Functional Workflow of the Surgical-YOLO System

3.1.4 Analysis of System Behavior and Operating Environment

All training was executed on a workstation running Windows 11, equipped with an NVIDIA RTX 3060 GPU (6 Gb VRAM), 16 GB of RAM, and an NVMe SSD for storing the dataset. The entire process utilized mixed precision training (AMP) on a single GPU which allowed peak memory usage to remain under the 6 GB VRAM limit. Batch size was equal to 6, in an effort to balance GPU memory consumption with the stability of the gradients obtained, at an input resolution of 640×640.

The Ultralytics YOLO Training Framework derived from Python, with the PyTorch deep learning engine; all models were pretrained from ImageNet weights, allowing for the use of a transfer learning from a large scale visual prior to assist in rapidly converging on a small spinal endoscopy dataset [42]. The optimizer is configured as 'auto', which Ultralytics resolves to SGD for nano-scale models. A constant learning rate was used throughout training to eliminate learning rate schedule as a confounding variable across configurations, effectively applying after the warmup period, as cosine scheduling is disabled in favor of the default linear decay. Training augmentation was configured to reflect realistic surgical imaging variations; the full parameter specification is documented in Chapter 4 Table 4-2.

Reproducibility is enforced through four-level deterministic seeding. Together these measures ensure that variation across the ten seeds for a given configuration reflects genuine training stochasticity, not hardware scheduling differences. The hardware constraint of 6 GB VRAM directly motivates the nano-scale model choice and batch size of 6 documented in Section 3.2.1; the strict reproducibility requirement motivates the four-level seeding protocol described above and implemented in Chapter 4 Section 4.4.1. One known exception: AdaptiveAvgPool2d, used inside the Coordinate Attention module, which has no deterministic CUDA backward implementation, so PyTorch emits a warning during CA runs. This has no effect on forward-pass reproducibility or the validity of reported mAP50-95 values.

3.1.5 Analysis of System Performance Requirements

The validation protocol used in Phase 2 used the basis of the statistical framework put forth in section 2.6.2 with the specified performance thresholds for acceptance criteria in this research. Coefficient of variation (CV) is the measure used to determine training stability. All configurations that are reported must achieve $CV < 1.5\%$ across ten seeds to be classified as stable. A paired t-test was used to evaluate statistical significance against the baseline distribution, with a p-value threshold of < 0.05 to establish significance. Cohen's d measures the residual effect size, with a threshold of $d > 0.8$ indicating that the improvement may be considered genuine and practically meaningful, rather than merely attributable to an artifact of sample size. A configuration that demonstrates all three of these criteria is thus established as a validated improvement.

Primary performance metric of the system will be measured by means of mAP50 to 95 (mean Average Precision over 10 different IoU threshold levels of .05 from .50 to .95). For single-position Phase 1 screening, a minimum improvement of approximately 1.0% in mean mAP50-95 over the baseline was used as a practical filter for candidate selection, on the basis

that improvements smaller than this threshold fall within the noise envelope of the three-seed screening protocol. No minimum latency target was specified. The attention modules employed in this work are parameter-light by design, and the nano-scale architecture is well within the real-time operating envelope on the hardware used in this work. These thresholds ($CV < 1.5\%$, $p < 0.05$, $d > 0.8$) are not arbitrary; they are set to ensure that comparisons between configurations differing by less than 1% in mAP50-95 are interpretable as genuine architectural differences rather than training noise, which is essential given the sub-percent magnitudes of some single-position ablation results.

3.2 System Design and Modeling

3.2.1 Overview of System Design

The design of Surgical-YOLO is organized around a single architectural principle: every design decision that is not the subject of experimental investigation must be held constant across all configurations, so that observed performance differences can be attributed unambiguously to the variable under study — the identity of the attention mechanism at each of the five positions in the feature pyramid. This principle will guide the development of the attention module wrapper and the selection of the training protocol and data augmentation.

The YOLO11n-seg will act as the baseline architecture for this research due to these three characteristics that meet the real-time constraints associated with surgical video segmentation: (1) The single-stage design permits deployment at frame rates suited to real-time surgical video; (2) The feature pyramid neck (FPN-PAN) produces representations across three spatial scales instantaneously, thus allowing the model to recognize large structures (e.g., intervertebral disc) down to small structures (e.g., herniated disk and nerve roots) in a single forward pass; and (3) The nano version has a parameter number of around 3.64 million and meets the limit on memory imposed by the RTX 3060 laptop GPU used in this study; thus, it is a realistic deployment target for hardware available in the operating room. The YOLO11n-seg configuration uses a depth multiplier of 0.50 and width multiplier of 0.25 on the nominal channel count specified in the YAML file, and all experimental variants of this architecture are modifications of it.

A C2MSCA module, which is a C2f-style wrapper that contains a Multi-Scale Cross-Axis Attention mechanism, is added to the standard YOLO11n-seg architecture as the laboratory baseline. MSCA will then be the baseline in which all experimental configurations will be compared to each other. In conducting system configuration experiments, another type

of attention mechanism will be substituted for the original Attention Mechanism at one or more of the five locations. The only independent measure that was isolated and incorporated into the respective C2f Style Wrapper was the type of Attention Mechanism used at the five pyramid locations.

Three design principles were utilized in developing the design to ensure reproducibility and fairness when conducting the ablation study. The first principle is Modularity; each attention module was developed independently as a class, which is registered in the Ultralytics task repository. As a result, to alter configurations, it is possible to replace modules without updating any code used for either training or evaluation in the YAML file. The second principle of fairness through isolation allows for every module to incorporate identical convolutional structures (beginning and ending convolutions). Therefore, the only differences between the performance metrics of the various attention mechanism types used and the performance metrics of the proposed models will be due to the type of attention mechanisms used and the locations in which they were used, not because of the way they are implemented. The third principle is Reproducibility; the four-level seeding protocol described in Section 4 provides a separate configuration/seed name for each training run and provides a method for independent verification of performance results.

The rationale for connecting the research hypothesis to the experimental design for this work is based on the hypothesis that each pyramid position will have a preference for a specific type of attention mechanism, and this requires that each attention mechanism be tested in isolation at each pyramid position. In order to meet these requirements for isolation, the development of a C2f wrapper allows for the maintenance of identical non-attention calculations across configurations. This research will identify the optimal attention mechanism for each pyramid structure. The chain of connection being referenced is represented by five acquisitions: - observed gap $\rightarrow$ position-level ablation $\rightarrow$ position-specific preferences $\rightarrow$ targeted hybrid design $\rightarrow$ statistical validation.

3.2.2 Design of System Data Architecture

The structure of the data architecture follows the standard structure of YOLO directories; there are images/train/, images/val/, labels/train/, labels/val/ and the data.yaml file containing the root directory path, path to each of the splits, number of classes (nc=6), and class names. Each labelled file has one entry for every instance in it (an integer class ID (0-5) along with x and y co-ordinates for polygon defined in normalized coordinates). This format is consumed directly by the Ultralytics data loader without additional preprocessing beyond the

augmentation pipeline described in Section 3.1.4. The dimensions for tensor shape at each of the attention positions in the nano scale configuration are listed in table 3-2.

Table 3-2 Tensor Dimensions at Each Attention Position in the Nano-Scale (n) Configuration

Layer	YAML Location	Channels (nominal)	Channels (nano $\times 0.25$)	Spatial Resolution	Functional Role
L11	Backbone terminus (after C2PSA, L10)	1024	256	20 $\times$ 20	Backbone consolidation before neck fusion begins
L15	FPN after P4 fusion (after C3k2, L14)	512	128	40 $\times$ 40	Feature preparation; output directly conditions downstream L19
L19	FPN output / P3 (after C3k2, L18)	256	64	80 $\times$ 80	High-resolution spatial output; direct input to Segment head
L23	PAN output / P4 (after C3k2, L22)	512	128	40 $\times$ 40	Mid-resolution balanced output; direct input to Segment head
L27	PAN output / P5 (after C3k2, L26)	1024	256	20 $\times$ 20	Low-resolution semantic output; direct input to Segment head

There is an input tensor having shape (B, 3, 640, 640), that gets processed in the Backbone layer progressively downsamples and increasing its channel depth and we get feature maps at each of the pyramidal level scales for P3 at stride 8 (80 x 80 spatial Dimensions), P4 at stride 16 (40 x 40 Dimensions), P5 at stride 32 (20 x 20 Dimensions) and then the feature maps are aggregated into output feature maps by processing the aggregate features across five attention locations through the FPN-PAN neck then creating 3 output feature maps that feed the Segment head. The head produces bounding box predictions, class logits, and mask coefficient vectors at each scale, which are combined with a learned prototype mask bank to yield per-instance binary segmentation masks.

3.2.3 Design of System Software Architecture

The Surgical-YOLO software architecture utilizes the main architecture of the Ultralytics YOLO framework as the basis for the following: Training Loop, Data Loading, Data Augmentation, Evaluation, and Model Serialization. The custom attention modules were developed by implementing two approaches: Python Module Registration and YAML Architecture Specification.

Custom attention module classes, i.e., MSCA, C2MSCA, TripletAttention, C2Triplet, CoordinateAttention, C2CA, ECA, C2ECA, are all created in Python source files and registered with Ultralytics using the tasks.py module resolution mechanism, which connects the module name strings in the YAML files with the appropriate Python class implementations. It is important to note that all custom C2f Wrapper classes provide the same constructor signature, making their use consistent with the Ultralytics YAML parsing convention. Therefore, any attention module may be changed by changing only one line in the YAML architecture file (specifically, the module name string located at the layer index). Furthermore, this change will not require changes to the corresponding training or evaluation code. Figure 3-4 depicts the software component relationship amongst the Ultralytics framework, the custom attention modules, architecture YAML files and the training script.

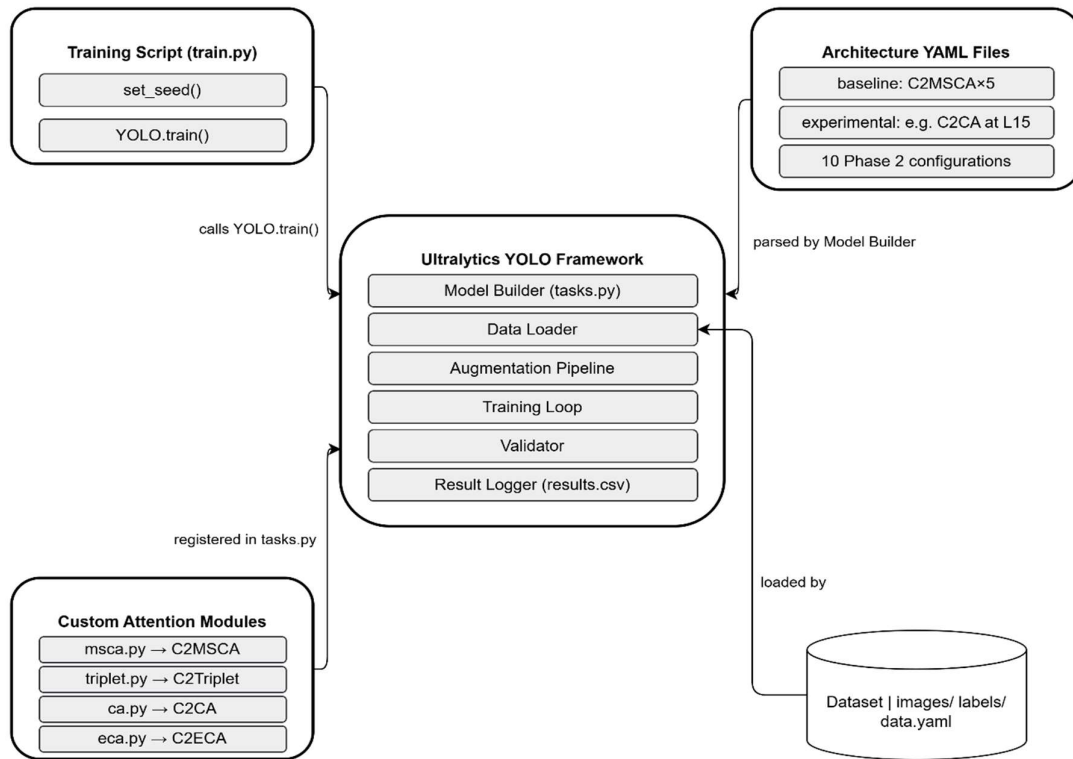


Figure 3-4 Software Component Diagram of the Surgical-YOLO System

Figure 3-5 provides complete details of the YOLO11n-seg architecture by providing all 5 attention positions. The diagram depicts the forward pass route from input to output, showing the backbone to top-down Path through FPN and bottom-up path through PAN to the Segment Head; indicating the specific attention module at each position, along with the channel count and spatial resolution at each level. The naming of each layer corresponds with the corresponding index of the YAML sequence.

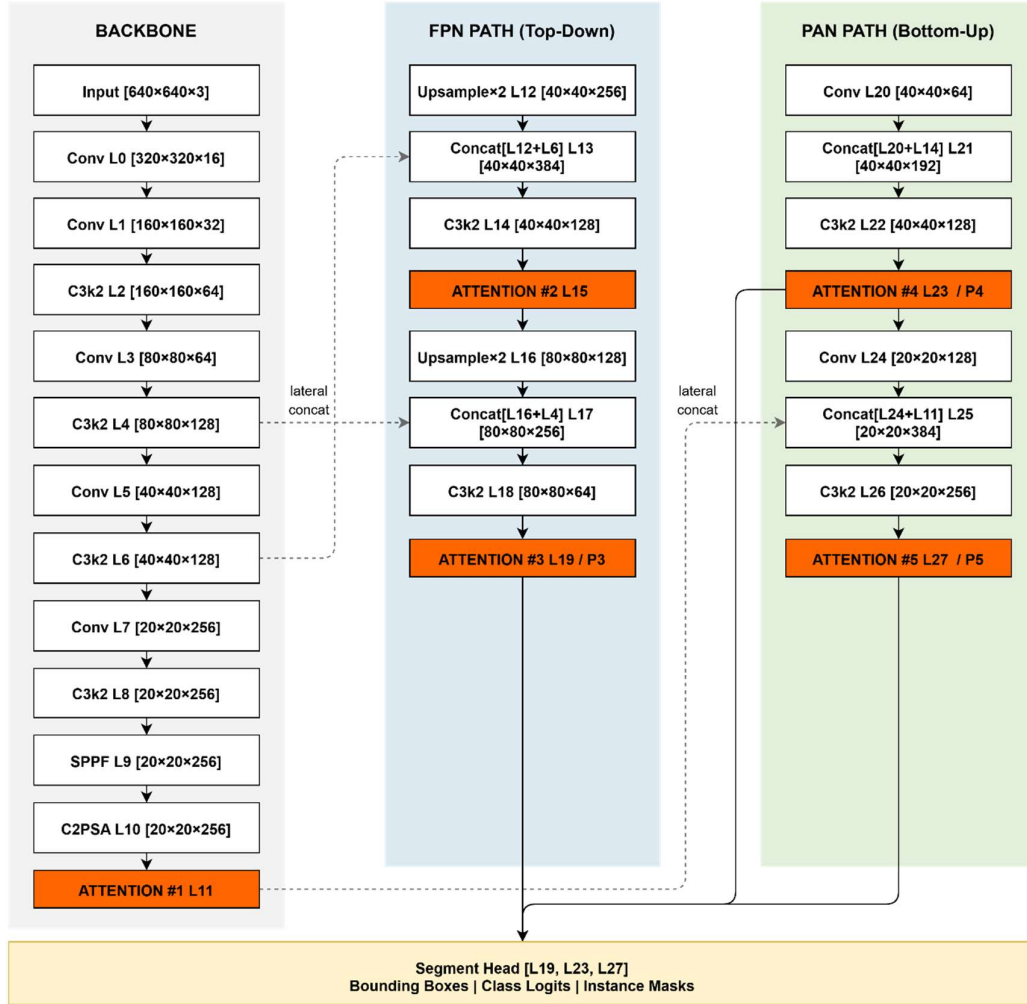


Figure 3-5 YOLO11n-seg Architecture with Five Attention Insertion Positions

3.2.4 Design of Experimental Interface

The training script and the associated collection of parameters in their YAML files form experimental interface for the system; the script will accept as input a dictionary of experiment names mapped to their respective YAML file locations and a list of seeds from which the script will create a list of configurations for the experiment. Each configuration will work through

each combination of configuration and seed value; thus, each training run will be independent from the others. Before loading the model, the script will set all random number seeds to the same specified number and will load the architecture, create a unique name for the run based on the experiment name and the respective index of the seed value used in this run, and provide all parameter values for the training process to the YOLO trainer. In this way, modifications to the configuration file do not require to modify the logic used to run the training or the evaluation code on the trained model, hence to add, remove, or modify the configuration, only need to modify the configuration dictionary.

The architecture for the model is defined in a YAML configuration file, with the layers being the primary experimental variable. Each YAML file provides a complete definition of the layer sequence in the model, with the locations of where to include the modules of each of the attention types (C2MSCA, C2Triplet, C2CA, C2ECA). The baseline YAML configuration file places C2MSCA at each positions uniformly. Each time, the experimental YAML modifies one or more of these entries. The number of configurations available for Phase 2 will be described in detail in Section 3.3.2.

3.2.5 Component-Level Design of the Attention Integration Framework

In Section 2.5, we described in detail the computational properties of MSCA, Triplet Attention, Coordinate Attention, ECA. Here we will describe the design logic of the C2f wrapper and we will describe the structural reasoning for placing each mechanism in the compound architecture.

All candidate attention mechanisms using a single C2f-style wrapper module. The input of this wrapper module has a number of parameters to allow for configurability: input channels (c_1), output channels (c_2), depth (n), and expansion ratio (default is 0.5). A 1×1 entry convolution cv_1 projects the input from c_1 to $2 \cdot c$ channels, where $c = \lfloor e \cdot c_2 \rfloor$, and splits the result into two tensors $y[0]$ and $y[1]$ of c channels each. $y[0]$ passes through unchanged as the first branch. $y[1]$ is also retained in the accumulation sequence and simultaneously passed through the attention block, whose output is appended as $y[2]$. If the wrapper depth n exceeds 1, each subsequent plain 3×3 convolution block operates on the last appended tensor and its output is appended in turn, forming an iterative transformation chain $y[0], y[1], y[2], \dots, y[n+1]$. All tensors in the sequence (initial split branches $y[0]$ and $y[1]$ and all subsequent block outputs) are then concatenated along the channel dimension, producing $(2+n) \cdot c$ channels total, which is projected back to c_2 channels by a 1×1 exit convolution cv_2 . An optional residual addition with

the input is applied when shortcut is enabled and the input and output channel counts are equal simultaneously. Figure 3-6 illustrates this wrapper architecture.

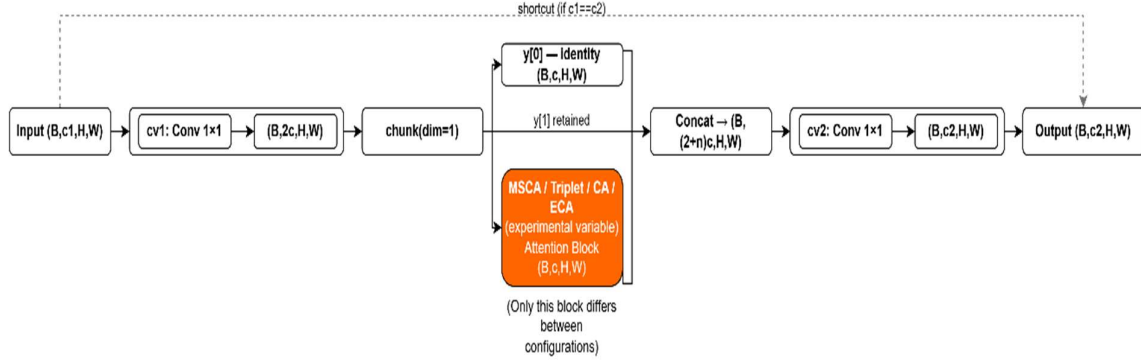


Figure 3-6 Uniform C2f Wrapper Architecture Shared by All Attention Module Variants

The wrapper design ensures experimental fairness in two respects. First, the entry and exit convolutions are structurally identical regardless of which attention mechanism occupies the interior block, meaning that any difference in forward-pass behavior between configurations can only originate from the attention block itself. Second, because all attention modules are dimensionality-preserving and the output shape equals the input shape (B, c, H, W), no reshaping or projection operations need to be added or removed when swapping mechanisms.

The positional assignment of each mechanism to its designated layer in the Hybrid-L15CA configuration is driven by the functional role of each position as described in Section 3.2.2 and elaborated in Section 3.3.1. At the two segmentation output positions L19 (80×80) and L23 (40×40), Triplet Attention is selected because its three parallel branches simultaneously capture $H \times W$ spatial, $C \times H$, and $C \times W$ cross-dimensional interactions. This design is well-matched to positions whose primary task is spatially precise boundary delineation at relatively high resolution. At the FPN fusion layer L15 (40×40), Coordinate Attention is selected because its directional one-dimensional pooling along height and width axes independently encodes the spatial relationship of merged features from two different backbone depths, providing a complementary inductive bias to the spatial cross-dimensional attention applied at the downstream L19. MSCA is retained at both L11 (backbone terminus) and L27 (P5 semantic output), both operating at 20×20 spatial resolution. L11 precedes all cross-scale fusion, so it does not have access to the multi-scale spatial context that makes Triplet effective at higher-resolution positions. At L27, the 20×20 spatial resolution suggests that spatially

selective attention is unlikely to resolve anatomical boundaries reliably; MSCA's multi-scale channel recalibration is better suited to semantic feature discrimination at this scale.

3.3 Detailed Design

3.3.1 Position-Aware Attention Allocation Design

As established in Chapter 1, the core hypothesis of this work is that different feature pyramid positions encode qualitatively different representations and therefore benefit from attention mechanisms with correspondingly different inductive biases. This portion of the report will go into detail about how we built the Hybrid-L15CA configuration and why we are assigning a certain attention mechanism to a particular position. The experimental evidence that validates these design choices is presented in Chapter 5.

As established in Section 3.2.2, the five positions occupy three functionally distinct roles: L11 consolidates backbone features before cross-scale fusion; L15 prepares fused features within the FPN path and is the only position whose output directly conditions a downstream attention layer (L19); and L19, L23, and L27 are direct segmentation head inputs at P3, P4, and P5 scales respectively.

Table 3-3 presents the final position-mechanism assignment of the Hybrid-L15CA configuration with the design rationale for each position.

Table 3-3 Hybrid-L15CA Position-Mechanism Assignment and Design Rationale

Layer	Resolution (nano)	Functional Role	Mechanism	Design Rationale
L11	20×20	Backbone terminus	MSCA	Consolidates backbone features before neck fusion. Spatial resolution is identical to L27 and too coarse for spatially selective attention. The backbone terminus precedes all cross-scale feature fusion; attention at this position operates on purely backbone-derived representations without the benefit of multi-scale context, and MSCA's multi-scale receptive field design is well-suited to consolidating these features.
L15	40×40	FPN fusion layer	CA	The only position that directly conditions a downstream attention layer (L19). Coordinate Attention's directional axial pooling encodes the spatial relationship of features merged from two backbone depths, providing a complementary inductive bias to the spatial cross-dimensional attention applied downstream at L19.
L19	80×80	P3 output / Seg. head input	Triplet	Highest-resolution position; primary role is fine-grained boundary delineation. Triplet Attention's simultaneous H×W, C×H, and C×W branches capture cross-dimensional spatial interactions well-suited to this requirement.
L23	40×40	P4 output / Seg. head input	Triplet	The mid-resolution feature representation at P4 balances spatial precision and semantic content. In this regime, cross-dimensional attention across all three axis pairs provides broader coverage than either purely spatial or purely channel-focused mechanisms.
L27	20×20	P5 output / Seg. head input	MSCA	Low-resolution semantic position. Spatial resolution is insufficient for spatially selective attention. Triplet Attention, Coordinate Attention, and ECA each have inductive biases oriented toward spatial selectivity or simple channel recalibration, neither of which matches the multi-scale semantic discrimination requirements of this position. Multi-scale channel recalibration is better suited to semantic feature discrimination.

3.3.2 Two-Phase Experimental Design

The experimental design proceeds in two phases with different precision-efficiency trade-offs. Phase 1 is a broad screening sweep designed to eliminate unpromising configurations at low computational cost. Phase 2 of the study will be a strict validation procedure to produce statistically reliable performance estimates of a set of curated configurations derived from Phase 1.

Phase 1 tests the performance of each type of attention mechanism independently (six non-MSCA candidates) on five positions. Each Phase 1 run trains for 50 epochs with 3 independent random seeds, generating position-mechanism preference signals across 66 training runs. From this evaluation process, a ranked preference map is created which shows the best-performing attention mechanisms for each of the five positions, from which the most promising candidates for hybridization are selected. One outcome of Phase 1 is that a training epoch of 50 was too short for some configurations to converge, L27-Triplet appeared to deteriorate around the 50 epochs mark, but showed signs of stable convergence by the end of full training, hence the cautious training regime of 100 epochs being utilized in Phase 2. Prior to the main Phase 1 sweep, each of the six mechanisms was tested in a 30 epoch training cycle using a single seed to ensure they would not produce unstable training results when integrated with other mechanisms.

At the end of Phase 2, a total of 100 training runs was completed. Each configuration will be trained for 100 epochs using 10 independent random seeds. For each configuration, there will be 10 mAP50-95 values, per seed available for summarizing results into a mean $\pm$ standard deviation per seed and performing a paired t-test with the ten base mAP50-95 values. The effect size will be calculated using Cohen's d and the stability will be calculated using the coefficient of variation. Table 3-4 includes a matrix showing all 10 of the configurations tested in Phase 2 along with their mechanism assignments.

Table 3-4 Complete Phase 2 Experimental Matrix: Ten Configurations and Mechanism Assignments

Configuration	L11	L15	L19	L23	L27	Type	Positions Modified
Baseline	MSCA	MSCA	MSCA	MSCA	MSCA	Uniform	0
L19-Triplet	MSCA	MSCA	Triplet	MSCA	MSCA	Single-position	1
L23-Triplet	MSCA	MSCA	MSCA	Triplet	MSCA	Single-position	1
L27-Triplet	MSCA	MSCA	MSCA	MSCA	Triplet	Single-position	1
Hybrid-MSCA	MSCA	MSCA	Triplet	Triplet	MSCA	Hybrid	2
Full Triplet	Triplet	Triplet	Triplet	Triplet	Triplet	Uniform	5
Hybrid- ECA	MSCA	MSCA	Triplet	Triplet	ECA	Hybrid	3
Hybrid-L27CA	MSCA	MSCA	Triplet	Triplet	CA	Hybrid	3
Hybrid-L15CA ★	MSCA	CA	Triplet	Triplet	MSCA	Hybrid (Optimal)	3
Full CA	CA	CA	CA	CA	CA	Uniform	5

Note: ★ marks the primary position-aware configuration. All ten configurations share identical training hyperparameters and only the attention mechanism YAML specification differs.

Hybrid - L15CA (★) modified 3 out of 5 of the positions, with L11 & L27 remaining the MSCA baseline. This selective 60% deployment profile is the most parsimonious in the Phase 2 set. The phase design ensures that the contribution of individual positions can be estimated by comparing single-position ablations (L19-Triplet, L23-Triplet, L27-Triplet) with their multi-position combinations (Hybrid-MSCA, Full Triplet), and that the benefit of adding L15-CA on top of the L19+L23-Triplet base (Hybrid-MSCA vs. Hybrid-L15CA) can be isolated. The uniform deployment baselines (Full Triplet, Full CA) provide upper-bound comparisons against which the efficiency of position-selective deployment can be assessed.

3.3.3 Training Configuration Design

All Phase 2 configurations share the training setup specified in Section 3.1.4 without exception. No hyperparameter, augmentation setting, or training callback was modified between configurations. The only variation across runs within Phase 2 is the random seed value and the architecture YAML file specifying the attention mechanism assignments.

Each configuration is trained with ten fixed random seeds, listed in Chapter 4 Table 4-3. The validation set is evaluated at the end of each epoch, and the best-epoch mAP50-95 across all 100 epochs is recorded from the results CSV as the performance estimate for that run. The ten best-epoch mean and standard deviation distributions used in the statistical analysis described in Section 3.1.5.

3.4 Summary of This Chapter

This chapter has outlined the overall analysis and design for Surgical-YOLO. In Section 3.1, we provided a description of the four requirements, characterizing the dataset, including the imbalance of six class annotations, with the Ligament and Nerve classes each being approximately 33%, and both Muscle and Herniation classes being critically underrepresented, at 2.4% and 4.9% respectively. Also described the deterministic split strategy to avoid any kind of data leaks between the training sets. A description of the operating conditions, the hardware used for training, an RTX 3060 Laptop GPU with 6 GB VRAM, along with all fixed hyperparameters for each of the configurations, has also been documented. Performance requirements were specified as $CV < 1.5\%$, $p < 0.05$, and Cohen's $d > 0.8$.

Section 3.2 developed the system design at three levels. The data architecture described the YOLO-format dataset structure and the tensor flow through the five attention positions, with precise nano-scale channel and spatial resolution specifications (Table 3-2). The software architecture identified the module registration mechanism enabling YAML-driven attention swapping and presented the full architecture diagram with all five positions labeled. The component-level design introduced the C2f wrapper as the uniform integration framework ensuring experimental fairness across all candidate mechanisms, and provided the positional design rationale for the Hybrid-L15CA assignment without anticipating the experimental results.

Section 3.3 provided the detailed design of the Hybrid-L15CA configuration (Table 3-3) with structural justification for each position-mechanism assignment based on functional role analysis, and presented the complete two-phase experimental matrix (Table 3-4) covering all

ten Phase 2 configurations. The training configuration was fully specified, including the ten-seed list and all hyperparameters. This design is specified at replication level: given the architecture YAML files and attention module implementations described here, the complete experimental program can be reproduced on equivalent hardware.

Chapter 4 Implementation

4.1 Implementation Environment and Tools

All experiments in this project were conducted on a single workstation whose specifications directly influenced every hyperparameter sensitive to memory and throughput constraints. This section documents the hardware and software environment to allow faithful reproduction of the results.

Table 4-1 Hardware and Software Environment

Component	Specification
GPU	NVIDIA GeForce RTX 3060 Laptop (6 GB GDDR6 VRAM)
CPU	Intel Core i7 (14-core, 2.3 GHz base)
System RAM	16 GB DDR5
Storage	NVMe SSD (dataset and checkpoint I/O)
Operating System	Windows 11 (64-bit)
Python	3.10.19
Deep Learning Framework	PyTorch 2.5.1 with CUDA 12.4
YOLO Framework	Ultralytics YOLO (v8.3.185 / YOLO11 branch)
Mixed Precision	PyTorch AMP (torch.cuda.amp)
Key Libraries	NumPy, OpenCV, PyYAML, Matplotlib, pandas

A batch size of 6 was used as documented in Section 3.1.4. Mixed-precision training (AMP) was enabled to reduce memory pressure and accelerate convolution operations, halving the memory footprint of activations during the forward pass. Ultralytics YOLO framework is the training and evaluation backbone for every experiment conducted in this thesis.

4.2 Dataset Preparation and Preprocessing

4.2.1 Raw Data and Annotation Format

The segmentData.py preprocessing pipeline (described in greater detail in Section 3.1.2) enables the conversion of raw LabelMe JSON annotations into YOLO-format label files through normalizing labels and converting coordinates, as well as the creation of deterministic training and validation datasets. The mapping from the class name includes all class name variations in Chinese characters to six standard English class names: IntervertebralDisc, Skeleton, Ligament, Muscle, Nerve and IntervertebralDiscHerniation.

4.2.2 Dataset Statistics

As illustrated in Figure 4-1, the training splits label distributions show an imbalance in classes, as referenced in Chapter 3, with relatively high counts of instances for Ligament and Nerve while there are very few instances of both Muscle and Herniation.

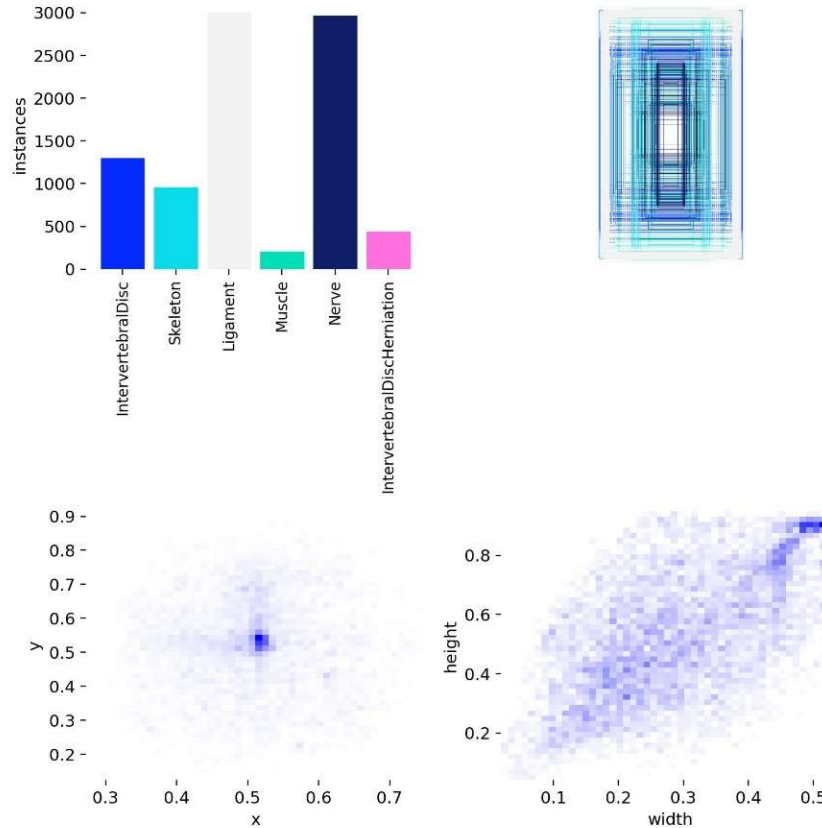


Figure 4-1 Upper left : Dataset label distribution showing instance counts per class. Lower left: Bounding box center coordinates. Lower right: Bounding box width–height scatter.

The bounding box center distribution (Figure 4-1, lower left) shows a strong concentration near image center ($x \approx 0.5$, $y \approx 0.5$), which is consistent with the circular endoscopic field of view framed against a black surround. The width/height scatter (Figure 4-1 lower right) shows a large distribution of instances on a diagonal line, which indicates that the vast majority of anatomical instances in the dataset have aspect ratios close to one and a wide range of surface area – from small nerve fibers to much larger ligament sheets. The need for three-level feature pyramid in YOLO11n-seg is driven by the range of scales.

4.2.3 Training Augmentation

Ultralytics performed online augmentation only via their data loader during training; no offline augmentation took place. The complete augmentation parameters used can be found in the experiment “args.yaml” file, as shown in Table 4-2.

Table 4-2 Online Training Augmentation Parameters

Parameter	Value	Effect
hsv_h	0.015	Hue jitter ($\pm 1.5\%$ of hue wheel)
hsv_s	0.7	Saturation jitter ($\pm 70\%$)
hsv_v	0.4	Value/brightness jitter ($\pm 40\%$)
fliplr	0.5	Horizontal flip (50% probability)
scale	0.5	Random scale jitter ($\pm 50\%$)
translate	0.1	Random translation ($\pm 10\%$ of image size)
mosaic	1.0	Mosaic composition (four-image grid, disabled in final 10 epochs)
erasing	0.4	Random rectangular erasing (40% probability)
auto_augment	randaugment	RandAugment policy applied after geometric transforms [43]
degrees / shear	0.0	Rotation and shear disabled
mixup / cutmix	0.0	Mixing-based augmentation disabled

Mosaic augmentation was turned off in the final ten epochs by the `close_mosaic: 10` parameter [44]; all rotation and shear augmentations were not applied due to the fact that endoscopic instrument movements do not provide arbitrary angles, and that strong geometric transforms will distort very long, thin polygons.

4.3 Attention Module Implementation

4.3.1 Registration with the Ultralytics Framework

For the experimentation described in this paper, the following custom attention module classes were created for use: MSCA, Triplet, CA, ECA, GC, SimAM, and Biformer. Each of these was registered within the Ultralytics framework by writing them into the Ultralytics module dispatch table via the `ultralytics/nn/tasks.py`. The modules can then be referenced in the YAML layer definition as if they were native Ultralytics modules, and they are automatically instantiated by the framework when the model is created.

4.3.2 Triplet Attention Implementation

Triplet Attention captures cross-dimensional interactions through three parallel branches targeting the $H \times W$, $C \times H$, and $C \times W$ axis pairs. The key implementation mechanism is tensor permutation: rather than building separate operators per axis pair, each branch permutes the input so that the desired axes occupy the spatial dimensions, applies an identical AttentionGate, then permutes back. AttentionGate compresses the permuted tensor along the channel dimension ($\text{dim}=1$) using ZPool, which concatenates channel-max and channel-mean into a two-channel descriptor. Then passes it through a Conv2d followed by BatchNorm2d and sigmoid activation, with no intermediate ReLU, since `relu=False` in the BasicConv call. Then multiplies the result element-wise with the branch input. The three branch outputs are averaged to form the final attended feature map. The implementation is shown below.

```
class TripletAttention(nn.Module):
    def __init__(self, no_spatial=False):
        super().__init__()
        self.no_spatial = no_spatial
        self.cw = AttentionGate()
        self.hc = AttentionGate()
        self.no_spatial = no_spatial
        if not no_spatial:
            self.hw = AttentionGate()
```

```
def forward(self, x):
    x1 = x.permute(0, 2, 1, 3)
    out1 = self.cw(x1).permute(0, 2, 1, 3)
    x2 = x.permute(0, 3, 2, 1)
    out2 = self.hc(x2).permute(0, 3, 2, 1)
    if not self.no_spatial:
        out3 = self.hw(x)
        return (out1 + out2 + out3) / 3
    return (out1 + out2) / 2

class AttentionGate(nn.Module):
    def __init__(self):
        super().__init__()
        k = 7
        self.pool = ZPool()
        self.conv = BasicConv(2, 1, k, p=k//2, relu=False)

    def forward(self, x):
        attn = torch.sigmoid(self.conv(self.pool(x)))
        return x * attn

class ZPool(nn.Module):
    def forward(self, x):
        max_pool = torch.max(x, dim=1, keepdim=True)[0]
        avg_pool = torch.mean(x, dim=1, keepdim=True)
        return torch.cat([max_pool, avg_pool], dim=1)
```

4.3.3 Coordinate Attention (CA) Implementation

Coordinate Attention decomposes spatial attention into two independent one-dimensional operations along the height and width axes. Two parallel adaptive average pooling operations of output sizes (H, 1) and (1, W) encode global context along each spatial axis into channel descriptors. The two descriptors are concatenated and jointly compressed by a shared 1×1 bottleneck followed by BatchNorm and SiLU, then split back into their height and width components. Each component is projected to the output channel dimension by its own 1×1 convolution, passed through sigmoid, and multiplied element-wise with the original input. The height and width gates applied independently.

```
class CoordinateAttention(nn.Module):
    def __init__(self, channels, reduction=32):
        super().__init__()
        self.pool_h = nn.AdaptiveAvgPool2d((None, 1))
        self.pool_w = nn.AdaptiveAvgPool2d((1, None))
        mip = max(8, channels // reduction)
        self.conv1 = nn.Conv2d(channels, mip, kernel_size=1)
        self.bn1 = nn.BatchNorm2d(mip)
        self.act = nn.SiLU(inplace=True)
        self.conv_h = nn.Conv2d(mip, channels, kernel_size=1)
        self.conv_w = nn.Conv2d(mip, channels, kernel_size=1)

    def forward(self, x):
        identity = x
        b, c, h, w = x.size()
        x_h = self.pool_h(x)
        x_w = self.pool_w(x)
        x_w = x_w.permute(0, 1, 3, 2)
        y = torch.cat([x_h, x_w], dim=2)
        y = self.act(self.bn1(self.conv1(y)))
        x_h, x_w = torch.split(y, [h, w], dim=2)
        x_w = x_w.permute(0, 1, 3, 2)
        a_h = self.conv_h(x_h).sigmoid()
        a_w = self.conv_w(x_w).sigmoid()
        return identity * a_h * a_w
```

CA is deployed at L15 in the Hybrid-L15CA configuration [45]. At this position the feature map is 512 channels at 40×40 spatial resolution. The reduction ratio of 32 yields a bottleneck width of 16 channels, keeping the parameter overhead negligible.

4.3.4 Efficient Channel Attention (ECA) Implementation

ECA performs channel recalibration without the dimensionality reduction bottleneck used by Squeeze-and-Excitation Networks. Global average pooling aggregates each channel to a scalar, producing a channel descriptor of length C. A one-dimensional convolution with an adaptively computed kernel size k then captures local cross-channel interactions, where k is determined from the channel count C via the formula below.

```
class ECA(nn.Module):
    def __init__(self, channels, gamma=2, b=1):
        super().__init__()
        t = int(abs((math.log2(channels) + b) / gamma))
        k = t if t % 2 else t + 1          # ensure odd kernel
        self.avg_pool = nn.AdaptiveAvgPool2d(1)
        self.conv      = nn.Conv1d(1, 1, kernel_size=k,
                                   padding=k // 2, bias=False)

    def forward(self, x):
        y = self.avg_pool(x)              # (N, C, 1, 1)
        y = self.conv(y.squeeze(-1).transpose(-1, -2))
        y = y.transpose(-1, -2).unsqueeze(-1)
        return x * y.sigmoid()
```

ECA was evaluated at the L27 position (P5, 20×20, 1024 channels) as a lightweight channel-only alternative to MSCA. The adaptive kernel formula yields $k=5$ for 1024 channels. The ECA and SE-Net are different in that SE-Net uses a dense layer to model the relationship of all pairs of channels, while ECA uses a 1D convolution with kernel size k to allow for only k neighbors to interact spatially. By only allowing limited interactions within each layer, ECA will have far fewer parameters than SE-Net for implementation but will maintain the same channel relationships.

4.3.5 Architecture YAML and Module Swap Mechanism

The configuration of all models is provided by one YAML file that describes the complete layer sequence of a given model. The Hybrid-L15CA Model (the main model type on which this work was focused) configures the input of an Attention position. The individual module identifiers for each attention position correspond to the module class name identified in the configuration YAML's Layer row.

```
# Hybrid-L15CA: position-aware attention configuration
# Backbone
backbone:
  - [-1, 1, Conv, [64, 3, 2]] # 0-P1/2
  - [-1, 1, Conv, [128, 3, 2]] # 1-P2/4
  - [-1, 2, C3k2, [256, False, 0.25]] # 2
  - [-1, 1, Conv, [256, 3, 2]] # 3-P3/8
```

```

- [-1, 2, C3k2, [512, False, 0.25]] # 4
- [-1, 1, Conv, [512, 3, 2]] # 5-P4/16
- [-1, 2, C3k2, [512, True]] # 6
- [-1, 1, Conv, [1024, 3, 2]] # 7-P5/32
- [-1, 2, C3k2, [1024, True]] # 8
- [-1, 1, SPPF, [1024, 5]] # 9
- [-1, 2, C2PSA, [1024]] # 10
- [-1, 1, C2MSCA, [1024, 1]] # 11 - Attention #1 (L11, MSCA)
head:
- [-1, 1, nn.Upsample, [None, 2, 'nearest']] # 12
- [[-1, 6], 1, Concat, [1]] # 13 cat
- [-1, 2, C3k2, [512, False]] # 14
- [-1, 1, C2CA, [512, 1]] # 15 - Attention #2
(L15, CA)
- [-1, 1, nn.Upsample, [None, 2, 'nearest']] # 16
- [[-1, 4], 1, Concat, [1]] # 17 cat
- [-1, 2, C3k2, [256, False]] # 18
- [-1, 1, C2Triplet, [256, 1]] # 19 - Attention #3
(L19, Triplet)
- [-1, 1, Conv, [256, 3, 2]] # 20
- [[-1,14], 1, Concat, [1]] # 21 cat
- [-1, 2, C3k2, [512, False]] # 22
- [-1, 1, C2Triplet, [512, 1]] # 23 - Attention #4
(L23, Triplet)
- [-1, 1, Conv, [512, 3, 2]] # 24
- [[-1,11], 1, Concat, [1]] # 25 cat
- [-1, 2, C3k2, [1024, True]] # 26
- [-1, 1, C2MSCA, [1024, 1]] # 27 - Attention #5
(L27, MSCA)
- [[19,23,27], 1, Segment, [nc, 32, 256]] # Seg head (P3/P4/P5)

```

To move from one configuration type (for example, baseline model with MSCA on all Positions) to another (for example, Hybrid-L15CA with MSCA on all Positions), the only changes necessary are to update five module names. The remainder of the configuration (layer definitions, Channel Counts, etc.) remains unaffected. By using the YAML swap mechanism, every configuration of the ten experiments conducted in this work (Section 4.5) can be implemented without a change or adjustments to any training code, thus eliminating any potential incorrect advantages to be gained by one of the configurations.

4.4 Multi-Seed Training Pipeline

4.4.1 Deterministic Training Setup

This program requires that any difference in performance observed between configurations is due to the specific architecture and not due to random variations of weight initialization or data ordering. The following four levels of determinism will be implemented for each training run.

- (1) `torch.manual_seed(seed)` and `torch.cuda.manual_seed(seed)` will be executed prior to creating any model or loading any training data to fix the random number generator state for both CPU and GPU implementations of the PyTorch library.
- (2) `numpy.random.seed(seed)` and `random.seed(seed)` are set to fix the NumPy and Python standard library random generators, which affect data augmentation decisions inside the Ultralytics data loader.
- (3) `torch.backends.cudnn.deterministic = True` forces cuDNN to use only deterministic convolution algorithms. The tradeoff is a slight drop in training speed, since cuDNN can no longer pick the fastest available kernel.
- (4) `torch.backends.cudnn.benchmark = False` prevents cuDNN from performing auto-tuning of convolution algorithms across batches, which would introduce run-dependent variation in the forward pass.

The CA module triggers a PyTorch warning on some seeds (adaptive_avg_pool2d_backward_cuda does not have a deterministic implementation) but this is purely cosmetic. It has no effect on the forward pass, the loss computation, or the weight updates.

4.4.2 Training Script

A single Python training script drives all Phase 2 experiments. The script iterates over a dictionary mapping experiment names to YAML paths and a fixed list of ten random seeds, calling `set_seed` before each model construction and invoking `model.train` with consistent hyperparameters. The seed index is appended to the run name to allow unambiguous identification of each output directory.

```
from ultralytics import YOLO
import torch, numpy as np, random
```

```
def set_seed(seed):  
    torch.manual_seed(seed)  
    torch.cuda.manual_seed(seed)  
    np.random.seed(seed)  
    random.seed(seed)  
    torch.backends.cudnn.deterministic = True  
    torch.backends.cudnn.benchmark = False  
  
experiments = {  
    'yolo11n_seg_c2triplet_c2ca_15':  
        'experiments/hybrid/yolo11n-seg-L19_23-c2triplet-L15-c2ca.yaml'  
}  
seeds = [42, 123, 999, 555, 777, 888, 111, 222, 333, 444]  
  
for exp_name, yaml_path in experiments.items():  
    for seed_idx, seed in enumerate(seeds, start=1):  
        set_seed(seed)  
        model = YOLO(yaml_path)  
        model.train(  
            data='data.yaml', epochs=100, imgsz=640,  
            batch=6, device=0, workers=8, amp=True,  
            seed=seed, deterministic=True,  
            project='runs/segment/hybrid',  
            name=f'{exp_name}_seed{seed_idx}'  
        )
```

The `set_seed` function is called before `YOLO(yaml_path)` rather than after, because the Ultralytics model constructor may itself call random operations during module weight initialization. Reproducibility was ensured by placing the seed list before construction.

4.4.3 Training Hyperparameters

The Phase 2 configurations for all experiments have the same hyperparameters. Therefore, all configurations had no tuning advantage. The hyperparameters used in all Phase 2 configurations were saved in the `args.yaml` file that Ultralytics created for each training run at the beginning of every training session.

Table 4-3 Phase 2 Training Hyperparameters (constant across all configurations)

Hyperparameter	Value	Rationale
Epochs	100	Ensures full convergence; 50-epoch Phase 1 data showed incomplete training
Batch size	6	Conservative fit within available VRAM budget with AMP enabled
Image size	640×640	YOLO11 standard input resolution
Optimizer	Auto (SGD)	Ultralytics auto-selects SGD for this model scale
Initial LR (lr0)	0.01	Standard SGD learning rate for YOLO training
Final LR (lrf)	0.01	Effective constant learning rate; cosine scheduling disabled, lr0 and lrf are equal.
Momentum	0.937	Standard YOLO SGD momentum
Weight decay	0.0005	L2 regularization
Warmup epochs	3	Linear LR warmup to prevent early divergence
AMP	True	Mixed-precision training for memory efficiency
Pretrained weights	True	ImageNet-pretrained initialization
Patience	100	Effectively disables early stopping; full 100 epochs always run
close_mosaic	10	Mosaic augmentation disabled in final 10 epochs
IoU threshold	0.7	NMS IoU threshold for post-processing
Seeds	10 fixed seeds	[42, 123, 999, 555, 777, 888, 111, 222, 333, 444]

By setting the patience value to 100 (the same value as the epoch budget), it effectively prohibits early stopping. Therefore, all configurations received an identical sequence of gradient updates and thus removes this potentially confounding influence when making comparisons.

4.5 Experimental Configurations and Evaluation Protocol

4.5.1 Baselines and Configurations

The experimental program is structured around a single fixed baseline: the supervisor-provided YOLO11n-seg architecture with MSCA inserted at all five attention positions (L11, L15, L19, L23, L27). All ten Phase 2 configurations are compared against this baseline using the same training procedure, dataset, and evaluation metric. Table 4-4 presents the complete experimental matrix (reproduced from Chapter 3 for reference) evaluated in Phase 2.

Table 4-4 Complete Phase 2 Experimental Matrix (Reproduced from Chapter 3 Table 3-4)

Configuration	L11	L15	L19	L23	L27	Type
Baseline	MSCA	MSCA	MSCA	MSCA	MSCA	Uniform
L19-Triplet	MSCA	MSCA	Triplet	MSCA	MSCA	Single-pos
L23-Triplet	MSCA	MSCA	MSCA	Triplet	MSCA	Single-pos
L27-Triplet	MSCA	MSCA	MSCA	MSCA	Triplet	Single-pos
Hybrid-MSCA	MSCA	MSCA	Triplet	Triplet	MSCA	Hybrid
Hybrid-L15CA ★	MSCA	CA	Triplet	Triplet	MSCA	Hybrid
Hybrid-L27CA	MSCA	MSCA	Triplet	Triplet	CA	Hybrid
Hybrid-ECA	MSCA	MSCA	Triplet	Triplet	ECA	Hybrid
Full Triplet	Triplet	Triplet	Triplet	Triplet	Triplet	Uniform
Full CA	CA	CA	CA	CA	CA	Uniform

★ denotes the primary position-aware configuration evaluated in Phase 2. The experimental matrix is designed to test three types of claims. The three single-position ablations (L19-Triplet, L23-Triplet, L27-Triplet) isolate the contribution of each individual pyramid level to allow position-level attribution. The hybrid configurations test whether combining validated single-position improvements produces additive or diminishing gains, and whether L27 benefits from a channel-only alternative to MSCA. The two uniform deployment configurations (Full

Triplet, Full CA) provide the upper-bound reference for non-position-aware strategies against which the hybrid designs are compared.

4.5.2 Evaluation Metrics

Model performance is evaluated using mean Average Precision (mAP). For each class at a given IoU threshold, predictions are first ranked by confidence score. A precision-recall curve is swept over this ranked list, and the Average Precision (AP) for that class is computed as the interpolated area under the curve following the COCO evaluation protocol, which samples 101 recall points uniformly from 0 to 1. mAP50 averages AP across all six anatomical classes at a single IoU threshold of 0.50, and mAP50-95 averages AP across ten IoU thresholds from 0.50 to 0.95 in steps of 0.05, then averages across all six classes with equal class weighting.

The YOLO11n-seg head produces separate bounding-box and mask prediction branches, so both bounding-box AP (B) and mask AP (M) are computed independently. Mask AP evaluates predictions using IoU computed on the binary instance mask rather than the bounding box, making it a more direct measure of segmentation quality. Precision measures the proportion of predicted instances at a fixed confidence threshold that correspond to a ground-truth annotation above the IoU criterion. Recall measures out of all the actual positive cases, how many did the model found correctly.

4.5.3 Statistical Evaluation Framework

A statistical framework has been developed and implemented for all ten configurations for the evaluation reported in Chapter 2. The statistical evaluation framework is as follows: Mean $\pm$ Standard Deviation across ten seeds; Coefficient of Variation ($CV = \sigma/\mu \times 100\%$); Paired t-test versus the baseline; and, Effect Size using Cohen's d.

An independent test dataset was not used in this study. Rather, all Phase 2 evaluation estimates are reported on the validation split consisting of 1,085 frames. This split serves both as the primary model selection criteria (the best.pt checkpoint) and as the ultimate evaluation dataset for the experimental design. This design choice is a limitation of the experimental scope. Validation-based reporting is standard practice in surgical video segmentation research where dataset sizes are limited [13], but readers should interpret all reported mAP50-95 values as validation-set estimates rather than generalization bounds.

Per-structure analysis is conducted by extracting the per-class AP50-95 values from the Ultralytics evaluation output for each seed and computing the mean improvement over baseline

for each of the six anatomical classes. This structure-level breakdown identifies which anatomical structures benefit from each attention strategy and provides clinically relevant interpretability that the aggregate mAP metric cannot. Metrics were extracted programmatically by running `model.val()` on each seed's best.pt checkpoint, collecting per-class AP50-95 values from the Ultralytics validation results object and the saved results CSV.

All percentage improvement values ($\Delta\%$) reported in Chapter 5 are computed as $\Delta\% = (\mu_{\text{exp}} - \mu_{\text{base}}) / \mu_{\text{base}} \times 100$, where μ_{exp} and μ_{base} are the ten-seed mean mAP50-95 values for the experimental and baseline configurations respectively. A paired t-test has been used to assess statistical significance using the Mean Average Precision (mAP) 50-95 results across both the ten seeds when paired by seed index; the reported p-value represents the two-tailed probability of observing a mean difference at least as great as the measured value under the null hypothesis of equal means. Cohen's d is computed as $d = (\mu_{\text{exp}} - \mu_{\text{base}}) / s_p$, where $s_p = \sqrt{((\sigma^2_{\text{exp}} + \sigma^2_{\text{base}}) / 2)}$ is the pooled standard deviation of the two ten-seed distributions. The Coefficient of Variation (CV) was defined as $CV = \sigma / \mu \times 100\%$ in section 2.6.2. Per-structure AP50-95 values were obtained by running `model.val()` using each of the seeds' best checkpoints, and collecting the per-class AP results from the Ultralytics validation output object; the mean and standard deviation of the ten seeds were calculated for each class. Inference speed is measured by the per image (preprocessing + inference + postprocessing in milliseconds) reported by `model.val()` with a batch size of 1 using a fully warmed up GPU, and $\text{FPS} = 1000 / \text{total_ms}$; the Number of Parameters and GFLOPs produced by `model.info()` in Ultralytics.

4.6 System Operation

4.6.1 Training Behavior

As seen in Figure 4-2, the hybrid-L15CA configuration training/validation curves are representative of all configurations at seed 2 (the source for all output figures throughout this section). The decrease in training losses for all four losses (box, segmentation, classification, and DFL) occurs consistently over 100 epochs; the validation loss curves decrease over the same 100 epochs, although there may be minor fluctuations from epoch to epoch, and there are no indications that either the training or validation curve plateaus or becomes unstable before epoch 80. The convergence profile of this configuration is consistent with that of a model that has been properly initialized using the correct augmentation and is consistent with other configurations. The first 20 epochs show a steep decline for the training loss due to the rapid

adaptation of classification head weights and the attention weights, followed by a more moderate deceleration in the decline rate as the model refines its learning of finer spatial detail.

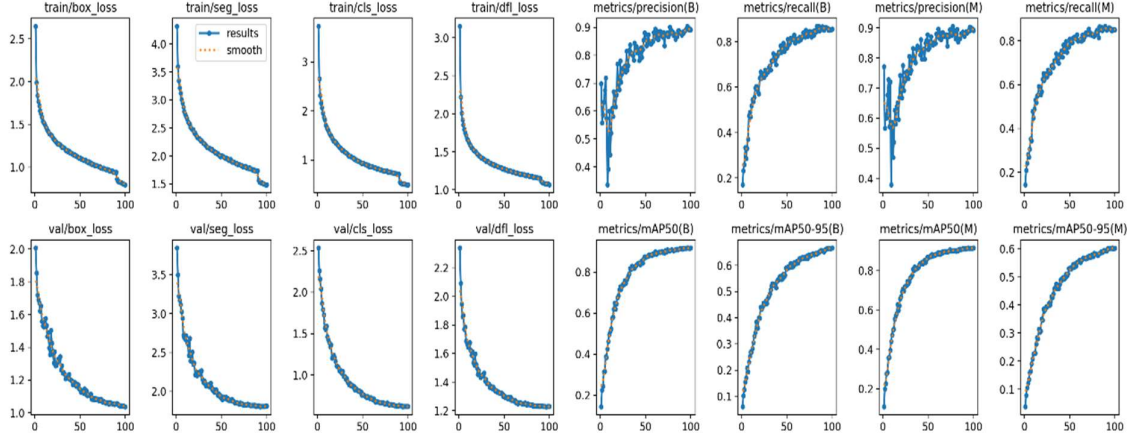


Figure 4-2 Upper row: Training loss curves (box, seg, cls, DFL losses). Lower row: validation metric curves (val losses and mAP50 / mAP50-95 for both box and mask predictions) over 100 epochs

The curve for one seed also indicates that the data from the one seed is representative of the data from the ten seeds because the coefficient of variation is very low across the seeds. The small separation between training and validation loss curves for the entire period of training indicates that the model generalizes well; furthermore, the data augmentation (used to prevent overfitting) should be successful despite the relatively small training dataset (5,873 training images). The precision and recall curves (metrics/precision(M) and metrics/recall(M)) indicate that the precision consistently exceeds 0.85 after approximately epoch 60, whereas the recall exceeds 0.85 at approximately epoch 71, remaining in the range of 0.84–0.87 during the last epochs of training. The results indicate a balanced trade-off between the number of false positive and missed detection classifications across the six anatomical classes.

4.6.2 Class-Level Segmentation Performance

Figure 4-3 shows the normalized confusion matrix for the Hybrid-L15CA configuration on the validation set, where rows are true classes, columns are predicted classes, and diagonal values are per-class recall.

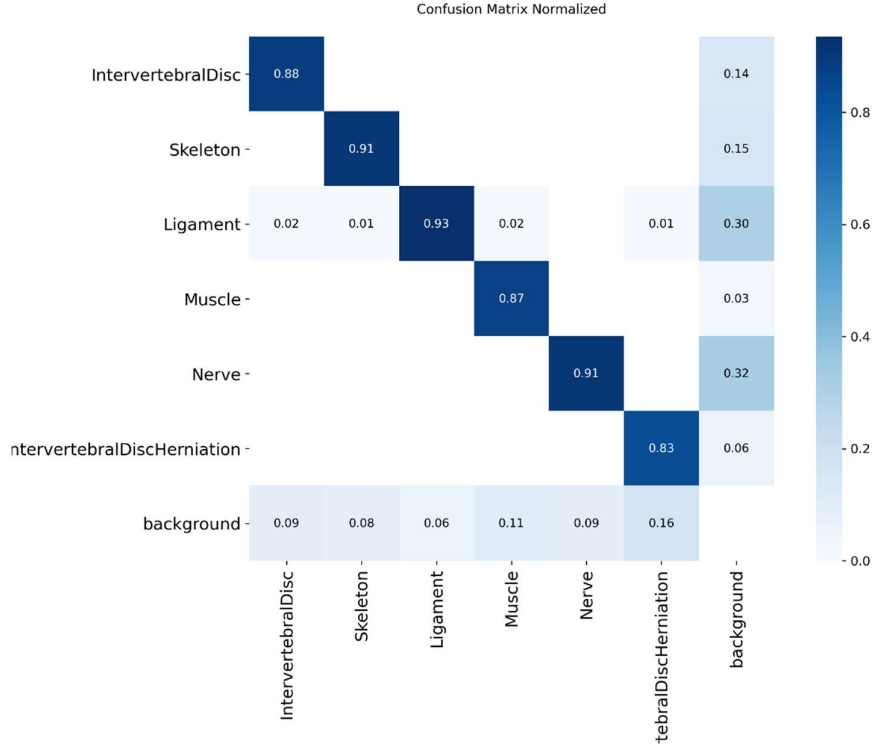


Figure 4-3 Normalized confusion matrix for Hybrid-L15CA on the validation set. Diagonal values represent per-class recall. The background column captures instances predicted as background (false negatives)

Ligament records the highest recall at 0.93; Skeleton and Nerve both follow at 0.91. These three classes also have the highest training instance counts. Intervertebral Disc achieves a recall of 0.88 and Muscle 0.87. IntervertebralDiscHerniation achieves the lowest recall at 0.83.

The most notable off-diagonal pattern is Ligament and Nerve both showing substantial prediction as background (0.30 and 0.32 respectively in the background column), which will be discussed in Chapter 5.

4.6.3 Inference Output Samples

Figure 4-4 presents a batch of twelve validation frames with predicted instance segmentation masks and bounding boxes overlaid. Each frame is drawn from Video5 of the clinical dataset and shows the endoscopic field of view, a circular region bounded by black, within which one or more anatomical structures are segmented.

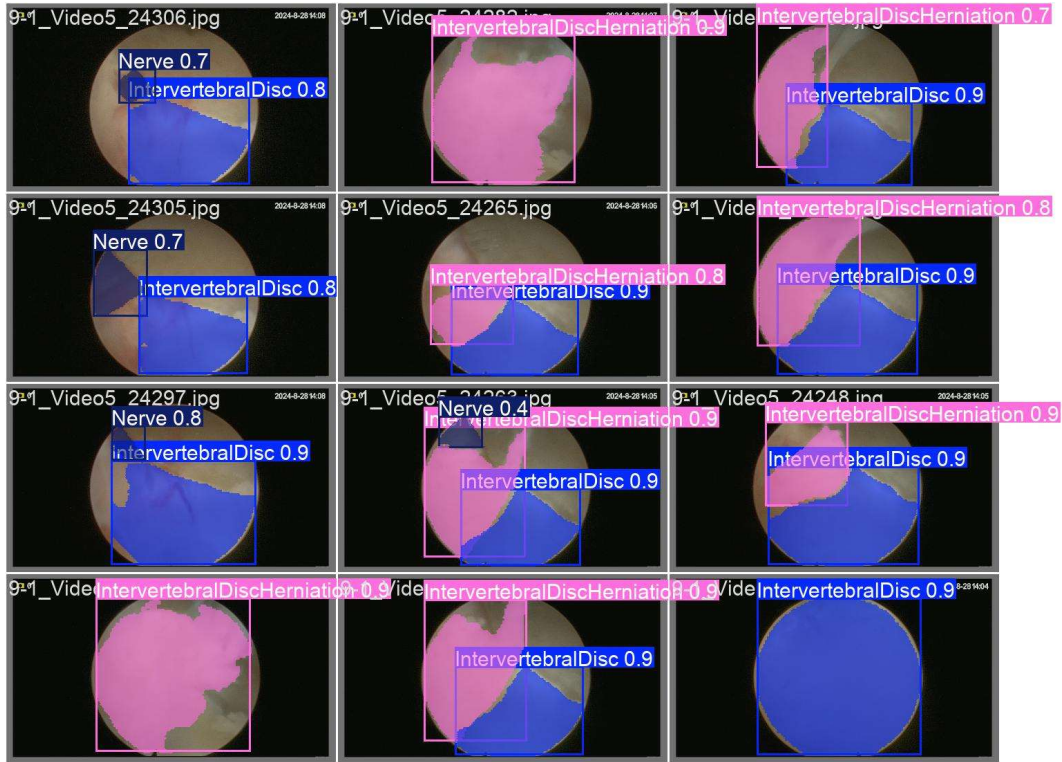


Figure 4-4 Representative validation batch with predicted instance segmentation masks.

IntervertebralDisc (blue) and IntervertebralDiscHerniation (pink) are consistently detected with confidence scores above 0.8. Nerve (dark) appears at the disc margin and is correctly localized in frames where it co-occurs with disc tissue

IntervertebralDisc detections score between 0.8 and 0.9 across the batch, with several IntervertebralDiscHerniation predictions reaching 0.9. Nerve predictions appear at confidence 0.7 in frames where the nerve is partially occluded by overlying disc tissue. The mask boundaries for disc and herniation tissue are sharp and closely follow the annotated polygon boundaries.

4.7 Summary of This Chapter

This chapter presented the complete implementation of the Surgical-YOLO experimental framework across four dimensions. Section 4.1 documented the hardware and software environment, establishing the hardware constraints that informed the batch size selection and the Ultralytics YOLO framework that provides the training infrastructure. Section 4.2 described the dataset preparation pipeline, from raw LabelMe JSON annotations through the segmentData.py normalization and conversion procedure to the final YOLO-format dataset of 6,958 frames, together with the online augmentation parameters applied uniformly across all experiments.

Section 4.3 provided the implementation of the attention mechanisms used in this work, with detailed code provided for the three non-MSCA mechanisms carried to Phase 2 rigorous validation (Triplet Attention, Coordinate Attention, and Efficient Channel Attention) including the full PyTorch source code for each, together with the YAML-based module swap mechanism that isolates architecture as the sole experimental variable. Section 4.4 described the multi-seed training pipeline, including the four-level deterministic reproducibility setup and the full hyperparameter table recorded from the args.yaml experiment logs. A critical software engineering design consideration throughout the implementation is a clear separation of the training and evaluation pipelines: best.pt checkpoint files are produced during training; the evaluation is performed in an independent call to model.val() on those checkpoint files without performing any augmentation. Separation of the training and evaluation pipeline ensures that the evaluation metric (mask mAP50-95 on the clean validation set) is not affected by augmentation decisions made during training and the evaluation metric can be reproduced for any reported result. Section 4.5 defined the guidelines for the evaluation protocol, including an experimental matrix of ten different configuration types, the primary evaluation metric being mask mAP50-95, and the three-level statistical report including mean $\pm$ std, coefficient of variation, and statistical significance from the t-test with effect size given by Cohen's d.

In section 4.6, representative output examples from the representative seed (seed 2) of the hybrid-L15CA configuration illustrated how the system trains: training convergence curves showed stable learning over 100 epochs for each of the four loss terms, a normalized confusion matrix showed per-class recall values along with expected background leakage for classes with large spatial extents, and visualizations of inference showed confidently predicted mask predictions for the critical disc and herniated discs. Overall, implementation details presented in this chapter contribute to validating the reproducibility and rigor of the experimental evaluations detailed in chapter 5.

Chapter 5 Experimental Results and Analysis

5.1 Test Environment

The workstation reported in Chapter 4 Table 4-1 was used to conduct the experiments for both Phase 1 and Phase 2 using the same PyTorch and Ultralytics software stacks that were described previously. Each configuration was trained and validated in the same environment, with identical hardware, identical software stacks, and identical hyperparameters. Therefore, the only difference in performance observed between each configuration resulted from the attention module configuration(s). The measurement of inference speed was performed by executing `model.val()` against the 1,085-frame validation set at a batch size of 1 and an input resolution of 640×640 when the GPU was fully warmed up. The reported frames per second (FPS) number for each configuration is equal to the total amount of latency (preprocessing + inference + postprocessing) for one image and the number of images processed based on the Ultralytics validation pipeline.

5.2 Test Program and Test Data Design

5.2.1 Phase 1 Screening Protocol

In Phase 1, a broad exploratory sweep was made to create position-mechanism preference signals at a low computational cost before carrying out a complete statistical analysis of each configuration. Six candidate mechanisms (Triplet Attention, Coordinate Attention (CA), Efficient Channel Attention (ECA), Global Context (GC), SimAM, and Biformer) were evaluated alone against the MSCA baseline using selected positions to yield a total of 22 single-position configurations tested for 50 epochs and 3 independent random seeds yielding a total of 66 training runs. Prior to Phase 1 screening, a preliminary compatibility check of 30 epochs and 1 seed per mechanism confirmed that all six candidate mechanisms could be incorporated within the architecture of the YOLO11n-seg without causing the system to exhibit any signs of instability during training. The primary output of Phase 1 is the preference map in Figure 5-1.

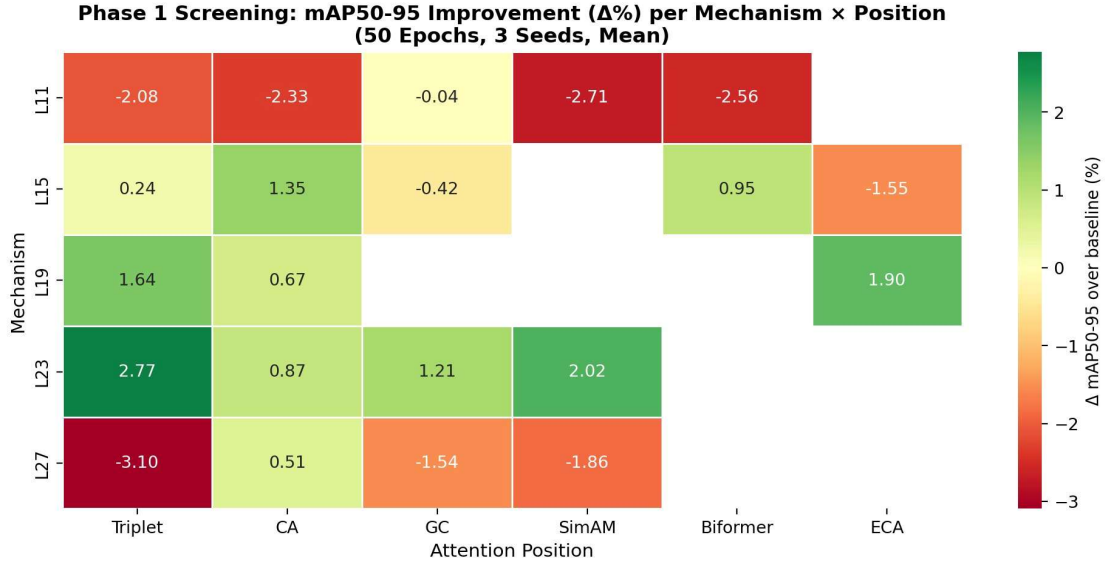


Figure 5-1 Phase 1 Screening Heatmap: mean mAP50-95 improvement ($\Delta\%$) over the 50-epoch MSCA baseline for each mechanism $\times$ position combination. Green cells indicate positive improvement and red cells indicate degradation. Empty cells mean configurations were not evaluated. The heatmap indicates that the highest positive signals for Triplet occur at L19 and L23 and CA occur at L15. In contrast, the majority of mechanisms assessed at L27 reveal either near-zero or negative outcomes. As such, for Phase 2, the decision-making process focused on Triplet (L19, L23) and CA (L15) as strong candidates, whilst eliminating all mechanisms assessed at L11 and considering the performance of GC, SimAM, and Biformer as weak options.

From an overarching perspective, there are distinct patterns emerging. All five attentions demonstrate an overall decline in performance when evaluated at L11: Triplet (-2.08%), CA (-2.33%), GC (-0.04%), SimAM (-2.71%), and Biformer (-2.56%). At the conclusion of the 50-epoch evaluation period, none of these five architectures was determined to provide any benefit to the performance of the backbone terminus. Conversely, among all five evaluated locations, L23 achieved the largest positive growth rate, with a +2.77% growth for the Triplet architecture and a +2.02% growth for SimAM. For single-position evaluations using CA, L15-CA had the highest improvement rate among all five evaluated locations, with a +1.35% growth rate. At the same time, L19-Triplet and L19-ECA produced a +1.64% and +1.90% increase, respectively. At L27, all architectures exhibit a reduction in performance but CA(+0.51%), with Triplet exhibiting the greatest reduction at -3.10%. GC, SimAM, and Biformer exhibited unreliable trends and were eliminated from Phase 2 evaluations.

During Phase 1, the evaluation strategy using 3-seed revealed a notable deficiency in variability across various configurations, with the variability level between 1.0% and 3.2% (refer to Figure 5-3, right-hand panel), which greatly exceeds the stability threshold of 1.5%

Moreover, the results for the L27-Triplet at an evaluation period of 50 epochs were reported to be -3.10% and L27-CA +0.51%, directly conflict with established validated outcomes from Phase 2 demonstrated that quantifying genuine advances over and above random training noise performance using these three 50-epoch seed estimates is not a reliable methodology. As such, the estimated outcomes led to a conservative approach for the ultimate goal of Phase 2, where a modified protocol of 100 epochs of evaluation time based upon 10 random seeds will be implemented.

5.2.2 Phase 2 Validation Protocol

The validation protocol for Phase 2 consisted of training 10 different selections over 100 training epochs with 10 independent random seeds each. Each evaluation session consisted of 100 separate training runs producing approximately 480 hours of computation time across GPUs. The selected configurations provided three different attributions for evaluation purposes. Single position ablations (Triplet L19, Triplet L23, Triplet L27) were developed to evaluate individual positions part-of-the-whole and support the testing of core hypothesis. Hybrid configurations (e.g., Hybrid MSCA, Hybrid ECA, Hybrid L27CA, Hybrid L15CA) were created to test whether the positions that generate ‘validated position’ improvements actually provide additive effects and whether alternative mechanisms at L27 outperformed the baseline performance. Uniform deployment experiments (Full Triplet, Full CA) provide upper-bound non-position-aware references.

Performance is estimated as the mean $\pm$ standard deviation of mask mAP50-95 obtained by re-validating each seed's best checkpoint (best.pt) on the held-out validation set using `model.val()` with no augmentation, which is the standard clean-evaluation protocol adopted in the literature. Statistical significance against the ten-seed baseline distribution uses a paired t-test ($p < 0.05$ threshold). Cohen's d provides the effect size, with $d > 0.8$ defined as a large effect. The coefficient of variation $CV = \sigma/\mu \times 100\%$ is the stability indicator, with $CV < 1.5\%$ required for a configuration to be considered reliably estimated. Per-structure AP50-95 improvements are computed by running `model.val()` on each seed's best checkpoint and averaging per-class results across seeds. All reported metrics are evaluated on the 1,085-frame validation split; no independent test set was held out, as described in Chapter 4 Section 4.5.3.

5.3 Test Run and Test Record

5.3.1 Baseline Performance

The MSCA baseline achieves mask mAP50-95 = 0.5751 ± 0.0038 across 10 seeds at 100 epochs, with CV = 0.66%, the lowest coefficient of variation in the entire experimental program and a reflection of MSCA's maturity as a well-tuned baseline configuration. This value is used as the base reference for all further comparisons in this research. An overview of the Phase 2 results for all ten configurations is shown in Figure 5-2. The specific results for each configuration will be analyzed further below.

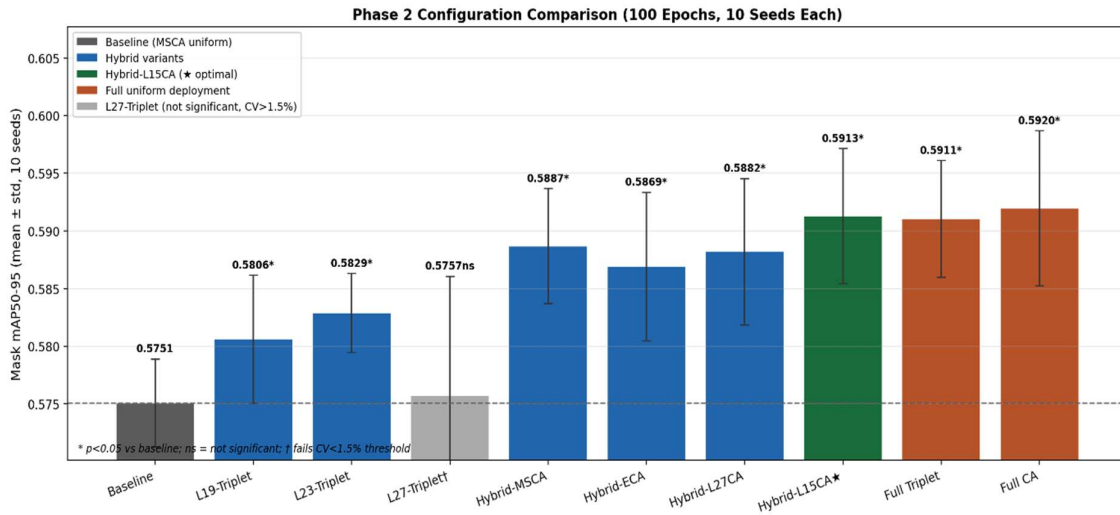


Figure 5-2 Phase 2 Configuration Comparison: mask mAP50-95 (mean ± std, 10 seeds, 100 epochs). Error bars show ±1 standard deviation. The dashed line marks the baseline (0.5751). Asterisks (*) denote statistical significance ($p < 0.05$) against the baseline. 'ns' means not significant. Hybrid-L15CA (★, green) achieves 0.5913, marginally crossing Full Triplet (0.5911) and within 0.0007 of Full CA (0.5920) with only 60% position deployment. L27-Triplet (grey) fails to show any significance and additionally exceeds the 1.80% CV that makes it the only configuration to fail the reliability threshold

5.3.2 Single-Position Ablations: Direct Test of the Core Hypothesis

The three single position Triplet test designs provide the clearest evidence of whether the effectiveness of an attention mechanism is dependent on the position within a pyramid. The diagrams in Table 5-1 and Figure 5-3 (left side) illustrate the results of the Triplet configuration experiments in which only one of the MSCA components were replaced by Triplet while all other MSCA components remained.

Table 5-1 Single-Position Triplet Ablation Results (Phase 2: 100 Epochs, 10 Seeds)

Configuration	mAP50-95	$\pm$ Std	$\Delta\%$	p-value	Cohen's d	Verdict
L19-Triplet (P3, 80×80)	0.5806	± 0.0055	+0.96%	0.0209	+1.163	Significant
L23-Triplet (P4, 40×40)	0.5829	± 0.0034	+1.36%	0.0014	+2.156	Significant
L27-Triplet (P5, 20×20)†	0.5757	± 0.0103	+0.11%	0.8309	—	Not Significant

Note: L27-Triplet fails the $CV < 1.5\%$ stability criterion ($CV = 1.80\%$) that confirm its result as unreliable.

Significance calculated by paired t-test against the 10-seed baseline distribution. L19 and L23 results confirm the position-dependence hypothesis. L27 fails both significance and the CV stability criterion, making it the only configuration in Phase 2 that cannot be reliably interpreted.

L19-Triplet (P3, 80×80) achieves 0.5806 ± 0.0055 , a +0.96% improvement ($p = 0.0209$, Cohen's $d = +1.163$). Statistical significance is demonstrated through the quantification of a large effect size. The highest definition of output in the pyramid is the 80×80 feature map at P3; detailed encoding of boundary data intended for use by the segmentation head. In addition, the targeted goal for spatial accuracy through the simultaneous modeling of $H \times W$, $C \times H$ and $C \times W$ cross-dimensional interaction with the Branch Point Connectivity of the Triplet Attention was to create sharper differentiation between the boundary areas of thin structures such as the nerve root and herniation margin.

L23-Triplet (P4, 40×40) produces the strongest single-position result: 0.5829 ± 0.0034 , a +1.36% gain ($p = 0.0014$, Cohen's $d = +2.156$). The high effect size and lowest CV of any modified configuration(0.59%) mean the positive effects of the hybrid configuration are very dependable and reproducible. The intermediate spatial-semantic balance for P4 makes it well suited for cross-dimensional attentional effects; it provides enough spatial precision for boundary discrimination and enough semantic richness to allow cross-channel spatial awareness for all three axis pairs at the same time.

L27-Triplet (P5, 20×20) achieves only 0.5757 ± 0.0103 , a nominal +0.11% change that is entirely non-significant ($p = 0.8309$). The empirical foundation of the position-awareness hypothesis is based on the findings of this study. Both P3 and P4's triplet attention shows an impressive increase in effectiveness as indicated by +0.96% for P3 and +1.36% for P4 but

provide little to no value at P5. There is also an excessive amount of CV (L27-Triplet's CV of 1.80%) above the 1.5% stability limit, indicating L27-Triplet is the only Phase 2 configuration that does not meet the reliability requirement. The combination of a null result with a high variance supports a misalignment of Triplet's spatially-selective inductive bias with the representational requirements of 20×20 semantic feature maps. At this level, each spatial attention element encompasses a 32×32 pixel space of the original input (a receptive field large enough to span an entire small anatomical feature) making it highly unlikely that spatially-selective mechanisms would improve boundary discrimination.

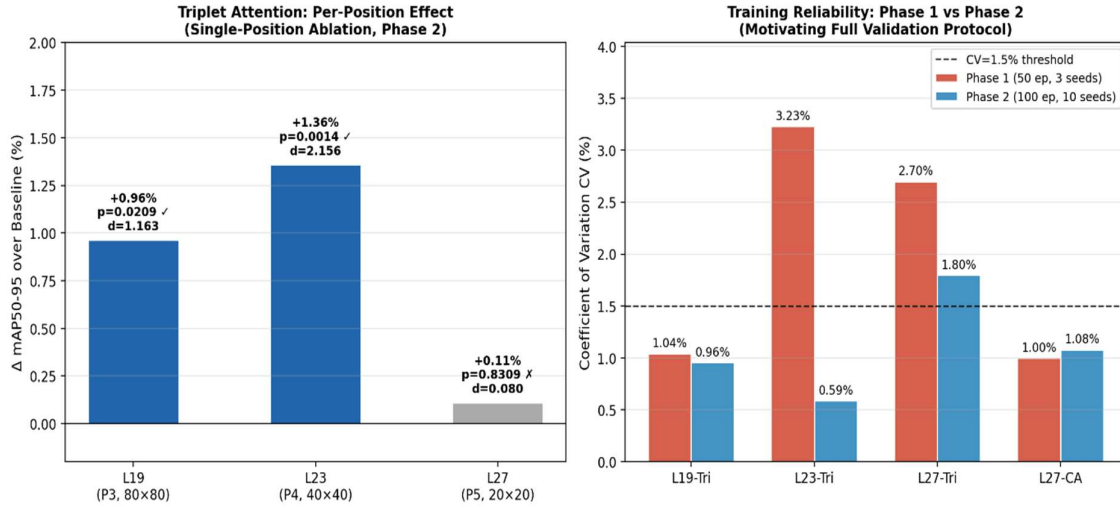


Figure 5-3 Left: Triplet Attention per-position effect (Δ mAP50-95 over baseline, Phase 2). The stark contrast between L19 and L23 (both significant) versus L27 (not significant) is the primary empirical validation of position-dependent attention effectiveness. **Right: Coefficient of variation at Phase 1 versus Phase 2 for matched configurations.** L23-Tri and L27-Tri show extreme Phase 1 CV values of 3.23% and 2.70% respectively, falling well above the dashed 1.5% reliability threshold. Phase 2 reduces all but L27-Triplet. CV values below 1.1% for these configurations demonstrate why 100-epoch, 10-seed validation is essential for reliable sub-percent mAP comparisons

5.3.3 Position Additivity: Hybrid-MSCA

Results from each configuration in this study (L19 and L23) confirm that these two performance gains are attributable to triplet attention; the hybrid-MSCA configuration (Triplet at L19 and L23, MSCA retained at L11, L15, L27) probes whether these performance improvements combine through additive means or metaphysically interact to create destructive interaction between the two performance sources. The hybrid-MSCA produced an overall result of 0.5887 ± 0.0050 , equating to a +2.36% relative improvement ($p < 0.0001$, Cohen's $d = +3.066$). The expected additive contribution for the combination of both individual position

performances would be +0.96% for P3 and +1.36% for P4, leading to a +2.32%; the actual benefit of +2.36% exceeded that predicted by 1.7%. This near-perfect additivity indicates that triplet attention at positions L19 and L23 provides complementary rather than redundant benefit, which aligns with the qualitative differences in the encoding of spatial frequency bands between P3 and P4 (fine-grained boundary detail at L19 and balanced mid-level spatial-semantic features at L23).

5.3.4 L27 Alternative Mechanisms

Because of the shortcomings of Triplet Attention in L27, two different channel-based methods were looked at to see if a different design than MSCA would help the P5 semantic location. The first of these two alternatives is Hybrid-ECA, which is on L27 where ECA is being substituted for MSCA (Triplet was still used on L19 and L23), and Hybrid-L27CA, which again replaces MSCA with CA only at L27 and under the same circumstances as Hybrid-ECA. The results of both Hybrid-ECA and Hybrid-L27CA are shown in Table 5-2.

Table 5-2 L27 Alternative Mechanism Results (Phase 2: 100 Epochs, 10 Seeds)

Configuration	L27 Module	mAP50-95	$\Delta\%$	p-value	vs Hybrid-MSCA
Hybrid-MSCA (reference)	MSCA	0.5887	+2.36%	<0.0001	—
Hybrid-ECA	ECA	0.5869	+2.06%	0.0005	−0.18% (worse)
Hybrid-L27CA	CA	0.5882	+2.28%	0.0001	−0.05% (worse)

Note: Hybrid-MSCA serves as the reference, isolating the L27 contribution by keeping L19 and L23 fixed at Triplet across all three rows.

Hybrid-ECA is inferior to Hybrid-MSCA, earning a score of 0.5869, and Hybrid-L27CA is almost as poor at 0.5882. Although the difference between these alternatives and Hybrid-MSCA might seem small, it's consistently lower than Hybrid-MSCA across all seeds. Under this condition of L27 with 20×20 spatial resolution, ECA does not afford the same capacity to capture multi-scale semantic relationships through 1D cross-channel interactions at a local level, nor does CA (directional axial pooling). Looking specifically at the result for CA, Phase 1 indicated that L27-CA performed at +0.51%. This would seem to indicate a strong result that

may have suggested some future promise; however, given the results from Phase 2 at 100 epochs with 10 seeds produced -0.05% to Hybrid-MSCA, we can conclude that the unreliability of Phase 1 is evident and that MSCA remains the superior baseline mechanism for L27.

5.3.5 Hybrid-L15CA: The Optimal Position-Aware Configuration

The rationale for Hybrid-L15CA is based on the strong Phase 1 single position performance data (L15-CA +1.35% is the largest $\Delta\%$ of any combination in the heatmap, see Figure 5-1). L15 has a unique structural position in that it is the only one with direct downstream effects on L19: channel attention at L15 performs directional semantic weighting before Triplet Attention at L19 applies cross-dimensional spatial gating. Therefore, it is a candidate for a sequential L15-CA type configuration with L15 having CA (as opposed to L19-L23, Triplet having L19-L23, and MSCA remaining on L11/L27).

Table 5-3 Complete Phase 2 Results Ranked by mAP50-95 (100 Epochs, 10 Seeds)

Configuration	mAP50-95	$\pm$ Std	Δ Baseline	p-value	Cohen's d	CV%	Sig. Struct.
Full CA	0.5920	± 0.0067	+2.94%	<0.0001	+3.093	1.13%	5/6
Hybrid-L15CA ★	0.5913	± 0.0059	+2.82%	<0.0001	+3.278	0.99%	5/6
Full Triplet	0.5911	± 0.0051	+2.77%	<0.0001	+3.565	0.86%	5/6
Hybrid-MSCA	0.5887	± 0.0050	+2.36%	<0.0001	+3.066	0.85%	—
Hybrid-L27CA	0.5882	± 0.0063	+2.28%	0.0001	+2.505	1.08%	—
Hybrid-ECA	0.5869	± 0.0064	+2.06%	0.0005	+2.235	1.10%	—
L23-Triplet	0.5829	± 0.0034	+1.36%	0.0014	+2.156	0.59%	—
L19-Triplet	0.5806	± 0.0055	+0.96%	0.0209	+1.163	0.96%	—
L27-Triplet†	0.5757	± 0.0103	+0.11%	0.8309	—	1.80%	—
Baseline	0.5751	± 0.0038	—	—	—	0.66%	—

Note: ★Hybrid-L15CA: the primary position-aware configuration. Green shading means recommended configuration. L27-Triplet additionally fails $CV < 1.5\%$ ($CV = 1.80\%$). Sig. Struct. means number of anatomical structures with statistically significant mAP50-95 improvement ($p < 0.05$) over the baseline. Per-structure analysis was not computed for Hybrid-MSCA and below-ranked configurations.

Hybrid-L15CA achieves mask mAP50-95 = 0.5913 ± 0.0059 , a +2.82% improvement over the baseline ($p < 0.0001$, Cohen's $d = +3.278$). L15CA does exceed Full Triplet (0.5911) by 0.0002 mAP points while changing only 3 out of 5 positions. Furthermore, the Full CA configuration exceeds L15CA by 0.0007 mAP points, however, the difference is less than the standard deviation of either configuration and well within the margins of noise for both. In Section 5.4, the implications of the per-structure and robustness of this finding are discussed extensively.

5.3.6 Uniform Deployment Baselines

Full Triplet achieves 0.5911 ± 0.0051 , a +2.77% gain ($p < 0.0001$, Cohen's $d = +3.565$). Compared to the ablation configuration, the approximate portion of the +2.77% gain attributed to L19-L23 combined (as isolated by Hybrid-MSCA) is a +2.36% total gain, with the remaining contribution (+0.41%) from the other 3 positions. This confirms that although L27 Triplet provided only a +0.11% contribution, it combined with L11/L15 provides a combined value of approximately +0.30% which ultimately results in only a small increase in mAP (0.5911) when compared to the Hybrid-MSCA gain of 0.5913. Furthermore, Hybrid-L15CA achieved a slightly greater contribution (+2.82%) with fewer positions modified than Full Triplet (+2.77%). This confirms that the targeted CA addition at L15 is more valuable than adding Triplet at L11 and L27 simultaneously.

With an overall score of 0.5920 ± 0.0067 , Full CA achieved a +2.94% average gain ($p < 0.0001$, Cohen's $d = +3.093$) over the course of this investigation; this is the final maximum mAP in the study and is in line with previous comparisons to both Hybrid-L15CA and Full Triplet coefficients, the latter of which has a higher coefficient of variation (1.13%); thus indicating greater variability in run to run value for Full CA than Hybrid-L15CA and Full Triplet. This information is documented in detail in Section 5.4.2.

5.4 Test Result Analysis

5.4.1 Validation of the Core Hypothesis

The core hypothesis of this thesis, that different feature pyramid positions benefit from different attention mechanisms, has been strongly supported through all of the ablation results related to single positions. If attention effectiveness were position-independent, Triplet applied at L27 would produce a gain of comparable magnitude to L19 (+0.96%) and L23 (+1.36%). Instead, L27-Triplet produces +0.11% with $p = 0.8309$ and a CV that exceeds the reliability

threshold. The same mechanism, the same wrapper, the same training procedure, yet zero benefit at one position, large significant benefit at two others.

A mechanistic explanation consistent with representational theory is as follows. At L19 (80×80) and L23 (40×40), the feature maps encode spatially localized information where the distinction between anatomical boundaries and background is expressed through spatial activation patterns across height and width dimensions. Triplet Attention's three-branch design ($H \times W$, $C \times H$, $C \times W$) targets exactly this kind of cross-dimensional spatial variation. At L27 (20×20), each spatial position covers a 32×32 pixel region of the original 640×640 input, a receptive field large enough to span an entire small anatomical structure. At this scale, spatial attention is unlikely to resolve inter-class boundaries reliably; the results suggest that multi-scale channel discrimination, which MSCA provides, is better suited to this position than the spatial and cross-dimensional mechanisms tested.

The heat map for Phase 1 (Figure 5-1) gives more verification of the evidence of performance degradation caused by all methods at L11 (backbone terminus, 20×20). The best performing method at L15 is CA (+1.35%), which suggests that the directional channel weighting of the fusion layer is better suited for use than using spatial gating. At L23 Triplet's Phase 1 signal of +2.77% is the largest positive cell on the heat map. Since both L15 and L23 have similar consistent patterns of positive cells across two phases, with differing training budgets and seed counts, there is strong support for the notion that there are true position dependent preferences within this dataset and model family.

5.4.2 Hybrid-L15CA versus Full CA: Robustness versus Absolute Performance

Hybrid-L15CA does not match the performance of Full CA (0.5920), due to the 0.0007 mAP50-95 difference, so Hybrid-L15CA and Full CA can be further dissected to see how much better Hybrid-L15CA performs in terms of being more robust in comparison to Full CA at mAP50-95. In the table 5-4 and figure 5-4 in this section, the various statistical dimensions are shown in the multi-dimensional space providing performance comparisons for all four of the highest performing configurations.

Table 5-4 Statistical Robustness Comparison: Top Four Configurations

Config	mAP50-95	CV%	Cohen's d	Sig. Struct.	Deployment	Strategy
Hybrid-L15CA ★	0.5913	0.99%	+3.278	5/6	60% (3/5)	CA+Triplet+MSCA; position-aware
Full CA	0.5920	1.13%	+3.093	5/6	100% (5/5)	CA uniform
Full Triplet	0.5911	0.86%	+3.565	5/6	100% (5/5)	Triplet uniform
Hybrid-MSCA	0.5887	0.85%	+3.066	—	40% (2/5)	Triplet at L19, L23 only

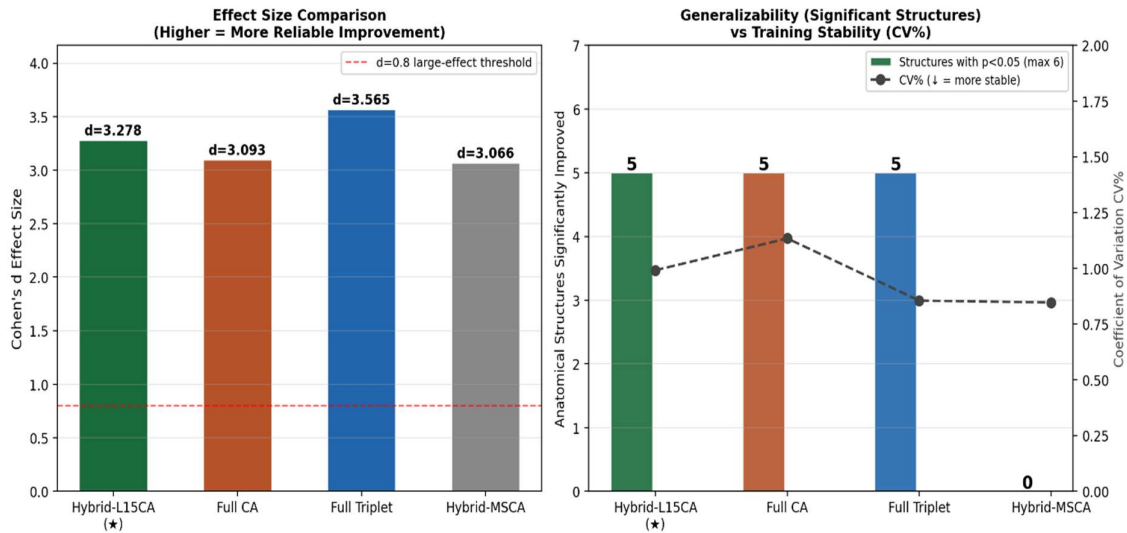


Figure 5-4 Left: Cohen's d effect size for the four top configurations. Full Triplet achieves the highest overall effect size ($d = 3.565$), followed by Hybrid-L15CA ($d = 3.278$) and Full CA ($d = 3.093$). All values exceed the $d > 0.8$ large-effect threshold by a substantial margin. Right: Number of anatomical structures with statistically significant improvement (bars, left axis) and training stability as CV% (line, right axis). All three top configurations achieve 5/6 significant structures. Hybrid-L15CA achieves the second-highest Cohen's d while maintaining the lowest CV among the three configs with mAP above 0.591 (0.99% vs Full CA 1.13%), providing a strong combination of effect reliability and training stability

Although Full CA achieved the highest overall mAP50-95 score of the three configurations (0.5920), the difference in performance was a mere 0.0007 mAP points from Hybrid-L15CA. All three top-performing configurations represented large improvements over five out of six anatomical structures with the only structure remaining non-significantly improved by any of the configurations being Muscle. In summary, therefore, the distinguishing

factor in Hybrid-L15CA vs Full CA is not the number or type of structures that are improved but rather the degree of improvement that occurs per anatomical position. Hybrid-L15CA's Cohen's d is greater than Full CA (Hybrid-L15CA = +3.278, Full CA = +3.093), and Hybrid-L15CA's CV (%) is lower than Full CA's (%). These advantages in training consistency persist despite Hybrid-L15CA modifying merely 3 out of the 5 positions of the five total positions of Full CA. With the 95.9% of the overall mAP gain achieved through Hybrid-L15CA's three modified positions, Hybrid-L15CA is a strong contender for further research, particularly at the limited deployment.

5.4.3 Per-Structure Analysis

The per-structure analysis is outlined in Table 5-5 and illustrated in Figure 5-5. This analysis provides a view of structural improvements that are hidden in the aggregate scores from mAP50-95.

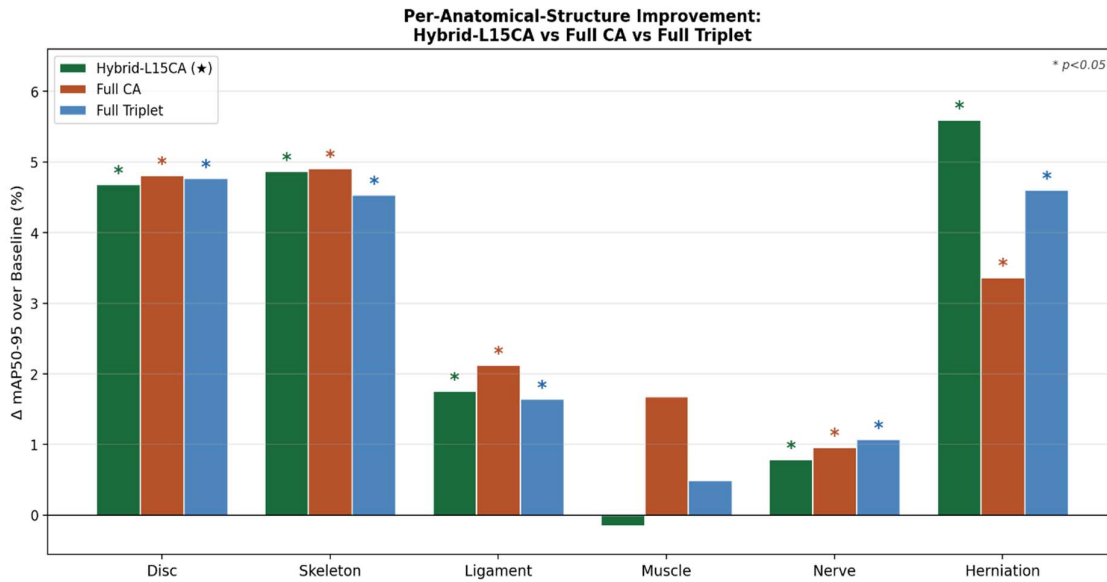


Figure 5-5 Per-anatomical-structure improvement (Δ mAP50-95 over baseline) for Hybrid-L15CA (green), Full CA (red), and Full Triplet (blue). Asterisks (*) indicate statistical significance ($p < 0.05$). All three configurations significantly improve 5/6 structures, with Muscle being non-significant across all. The key differentiator is Herniation: Hybrid-L15CA achieves +5.60% ($p = 0.0015$) larger than Full CA's +3.37% ($p = 0.0174$) and the largest absolute gain on this clinically critical small class across all configurations

Table 5-5 Per-Structure Improvement: Hybrid-L15CA, Full CA, and Full Triplet vs Baseline (*, $p < 0.05$; ns, not significant)

Structure	Δ L15CA	p	Sig	Δ Full CA	p	Sig	Δ Full Triplet	p	Sig
Disc	+4.68%	0.0001	*	+4.81%	<0.0001	*	+4.77%	<0.0001	*
Skeleton	+4.87%	<0.0001	*	+4.91%	<0.0001	*	+4.53%	<0.0001	*
Ligament	+1.76%	0.0073	*	+2.13%	0.0001	*	+1.64%	0.0004	*
Muscle	-0.15%	0.9074	ns	+1.68%	0.2159	ns	+0.49%	0.6584	ns
Nerve	+0.79%	0.0018	*	+0.96%	0.0427	*	+1.07%	0.0055	*
Herniation	+5.60%	0.0015	*	+3.37%	0.0174	*	+4.61%	0.0080	*

Note: Improvement values are mean Δ mAP50-95 averaged across 10 seeds relative to the mean baseline. p-values from paired t-test across the 10-seed per-class AP50-95 distributions. Exact per-seed per-class values were collected from model.val() on each seed's best checkpoint.

The best distinction between configurations can be seen with Herniation: Hybrid-L15CA at 5.60%, Full Triplet at 4.61% and Full CA at 3.37%; each of the three configurations has statistically significant results, but Hybrid-L15CA has gained 66.2% more than Full CA and is the best performing configuration for that class. The main clinical benefit of the position-aware design is that the sequential CA→Triplet refinement pipeline from L15 to L19 provides increased sensitivity to small, visually subtle herniation boundaries, while uniform CA deployment does not provide equivalent improvement. Disc, Skeleton and Ligament have substantially similar improvements from all three configuration types. All three structures benefit broadly from any multi-position attention and do not have enough distinguishing characteristics between configuration strategies.

Muscle is non-significant between any of the three configurations (Hybrid-L15CA -0.15%, $p=0.9074$; Full CA +1.68%, $p=0.2159$; Full Triplet +0.49%, $p=0.6584$) due to its extremely limited training examples (only 202 instances). Improvements for Nerve across all three configurations are low in absolute terms, with the smallest improvement Hybrid-L15CA (+0.79%, $p=0.0018$), however the gain remains significant based on a 10-seed paired t-test.

5.4.4 Computational Efficiency Analysis

The computational efficiency analysis is illustrated in Table 5-6 and Figure 5-6 with the inference speed results for four representative configurations. The most salient finding of this study is that all three arrangements of attention modules perform better than the MSCA baseline.

Table 5-6 Inference Speed Comparison (batch=1, 640×640, RTX 3060 Laptop GPU)

Config	Infer. (ms)	Postproc. (ms)	Total (ms)	FPS	Params	GFLOPs
Baseline	4.3	1.7	6.2	161.8	3.636M	11.0
Hybrid-L15CA★	3.5	1.5	5.2	191.6	3.547M	10.6
Full Triplet	3.2	1.3	4.7	213.7	3.259M	10.3
Full CA	3.1	1.3	4.6	219.8	3.268M	10.3

Note: Preprocessing and postprocessing times are included in total latency. $FPS = 1000 / total_ms$.

Measurements from model.val() with GPU fully warmed up over the 1,085-frame validation set. Speed values are averaged across all ten seeds for each configuration

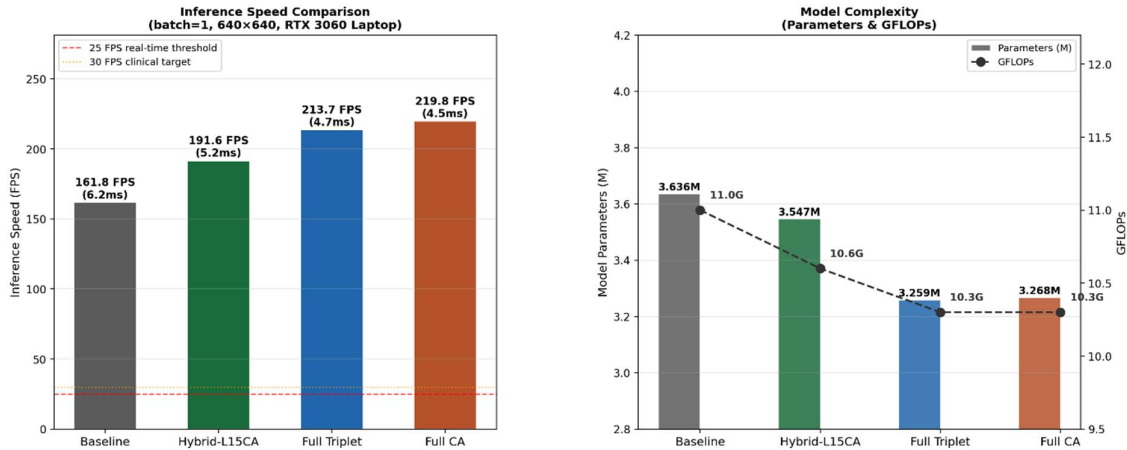


Figure 5-6 Left: Inference speed comparison for four representative configurations. All attention-modified architectures exceed 190 FPS, clearing the 25 FPS real-time threshold by more than 7×. Right: Model parameters and total GFLOPs. GFLOPs range from 10.3 (Full Triplet, Full CA) to 11.0 (Baseline). The lighter attention modules reduce both parameter count and compute relative to the MSCA baseline, with Hybrid-L15CA at 3.547M parameters and 10.6 GFLOPs representing a 2.4% parameter reduction while achieving the best position-aware accuracy

As shown in Table 5-6, all three attention-modified configurations substantially outpace the baseline (161.8 FPS), with Hybrid-L15CA achieving 191.6 FPS (an 18% improvement), and Full Triplet and Full CA reaching 213.7 and 219.8 FPS respectively. All four configurations comfortably exceed the 25 FPS minimum for real-time surgical video and the 30 FPS clinical target, with margins of $6.5\times$, $7.7\times$, $8.5\times$, and $8.8\times$ respectively. The baseline being slowest reflects a property of MSCA's multi-scale depthwise convolution frontend: the parallel 3×3 , 5×5 , and 7×7 depthwise branches introduce irregular memory access patterns that do not map efficiently to GPU warp execution, whereas Triplet Attention, CA, and ECA all use simple pooling and convolution operations that align well with GPU memory coalescing. This indicates that selecting Hybrid-L15CA as the model architecture over the MSCA baseline does not result in an inference speed decrease (average speed increase of 18%) and improved segmentation accuracy (average increase of +2.82% mAP50-95).

The mAP-per-position metric provides a measure for how much of the overall performance increase a given modified position contributes to on average. Thus, Hybrid-L23-Triplet configuration (only 1 of 5 positions modified) has an mAP-per-position of 1.36%, Hybrid-MSCA configuration (2 of 5 positions modified) has an mAP-per-position of 1.18%, Hybrid-L15CA configuration (3 of 5 positions modified) has an mAP-per-position of 0.94%, whereas both Full Triplet configuration and Full CA configuration (5 of 5 positions modified) have mAP-per-position scores of 0.56% and 0.59%, respectively. The diminishing return of performance per additional position modified is consistent with the representation overlap of modified positions; however, the Hybrid-L15CA was the only arrangement (0.94% per position) superior to both Full Triplet and Full CA arrangements (0.56% and 0.59%, respectively). As such, in this experimental context, Hybrid-L15CA benefits more from the targeted modified position as opposed to the uniform distribution of modified positions.

All attention modules in this work are parameter-minimal relative to the 3.636M-parameter YOLO11n-seg baseline, and total GFLOPs (backbone + neck + head + attention) vary between 10.3 (Full Triplet, Full CA) and 11.0 (Baseline). Hybrid-L15CA carries 3.547M parameters and 10.6 GFLOPs, representing a 2.4% parameter reduction and 0.4 GFLOPs reduction relative to the baseline. Full Triplet and Full CA reduce parameters further to 3.259M and 3.268M respectively, both at 10.3 GFLOPs. The lighter attention modules replace the more parameter-heavy MSCA wrappers and simplify the computation graph, which explains the counterintuitive finding that configurations with additional attention modifications are faster at

inference than the MSCA baseline. The maximum GFLOPs variation of 0.7 across all configurations is architecturally negligible.

5.4.5 Limitations and Scope

The results presented in this chapter establish that position-aware attention deployment is a viable and effective design strategy for YOLO-based spinal endoscopic video segmentation. Five limitations constrain the scope of these conclusions and should be considered when interpreting or extending this work.

First, this study's findings are based on a single clinical dataset consisting of 6,958 annotated frames collected from one institution and a single procedure type (percutaneous endoscopic spinal decompression), and thus cannot be considered to generalize to other anatomical regions, procedures (e.g., laparoscopic, robotic-assisted, colonoscopic), or to datasets with various annotation quality and class distributions. In addition, the relatively small size of the training dataset, especially for minority classes (e.g., Herniation, 434 instances; Muscle, 202 instances) affects confidence in the validation of position-level mechanism preferences for these classes specifically.

Second, the mechanism sets were limited to four established families: Triplet, CA, ECA, and MSCA baseline. All tested alternatives at L27 and L11 either matched or degraded baseline performance, but this does not rule out the existence of better-suited mechanisms at these positions among the broader attention literature. Deformable attention, window-based self-attention, or task-specific designs tailored to semantic feature maps were not evaluated, and the absence of improvement from the tested alternatives at L11 and L27 may reflect a limitation of the candidate set rather than a fundamental property of those positions.

Third, the inference speed measurements, parameter counts, and GFLOPs reported in Section 5.3.6 are hardware-specific to the RTX 3060 Laptop GPU and may not generalize uniformly across GPU architectures. The finding that MSCA is slower than lighter attention modules due to irregular memory access patterns from its multi-scale depthwise convolution frontend is real but hardware-dependent; on different memory hierarchies the relative speed ordering of configurations could differ. No attention map visualization was conducted to confirm that the position-specific mechanisms are attending to anatomically meaningful regions, leaving open the question of whether the observed mAP gains reflect the intended representational effects or arise from other computational interactions within the C2f wrapper.

Fourth, the system has not been validated in a live surgical deployment environment. All experiments were conducted offline on pre-recorded video frames. Factors including streaming latency, GPU availability in the operating theatre, calibration between the segmentation output and the physical instrument position, and clinical workflow integration have not been addressed. The inference speed results (Section 5.3.6) demonstrate that the model operates within the real-time throughput budget, but end-to-end deployment readiness has not been established.

Fifth, all conclusions regarding position-level mechanism preferences of this study are based on a single family of models (YOLO11n-seg) trained on a single dataset. The fact that Triplet scored higher than the other mechanisms on P3 and P4 but not on P5 reflects a dataset-specific finding that cannot be generalized to other FPN-based architectures or input resolutions or annotation schemes. The term 'position-aware' describes the design strategy validated in this work; it does not constitute a general theory of attention placement that has been validated across multiple settings.

5.5 Summary of This Chapter

This chapter presented the complete experimental results and analysis of the two-phase investigation into position-aware attention deployment for spinal endoscopic video segmentation.

Section 5.2 described the two-phase protocol. Phase 1 (50 epochs, 3 seeds, 66 runs) generated the position-mechanism preference heatmap from which Triplet Attention at L19/L23 and CA at L15 were identified as primary Phase 2 candidates, while simultaneously demonstrating its own unreliability through Phase 1 CV values of up to 3.23% and rank reversals that contradicted Phase 2 outcomes. Phase 2 (100 epochs, 10 seeds, 100 runs) delivered the statistically validated final results.

Section 5.3 presented all ten Phase 2 configurations against the MSCA baseline (0.5751 ± 0.0038 , CV 0.66%). Single-position ablations validated the core hypothesis through position-dependent Triplet results. Hybrid designs confirmed near-perfect additivity (101.7%) and MSCA optimality at L27. Hybrid-L15CA (0.5913) surpassed Full Triplet (0.5911) with fewer modifications and ran 18% faster than the MSCA baseline. Full numerical results are in Tables 5-1 through 5-6.

Section 5.4 analyzed these results across four dimensions. The core hypothesis is validated by the L27 null result. The Hybrid-L15CA versus Full CA comparison shows that a

0.0007 mAP gap, well within measurement noise, is accompanied by a 6.0% larger Cohen's d effect size, 12.4% lower training variance, and a 66.2% larger improvement on the clinically critical Herniation class. The per-structure analysis showed all three top configurations significantly improve 5/6 structures; the primary differentiator is Herniation magnitude, where Hybrid-L15CA's +5.60% substantially exceeds both Full CA (+3.37%) and Full Triplet (+4.61%). The efficiency analysis established that Hybrid-L15CA achieves 0.94% mAP gain per modified position, substantially exceeding Full Triplet (0.56%) and Full CA (0.59%). It also reduces parameter count by 2.4% and inference latency by 18% relative to the baseline.

Chapter 6 Work Summary and Reflection

6.1 Work Summary

In this research, we want to determine whether each of the five distinct attentional positions within the YOLO11N-seg feature pyramid has its own preferred way of representing information. In addition, we wanted to see if we could leverage this knowledge about positional preference distributions via position-specific mechanism to achieve the same or better performance than using a uniform deployment of attention on a real-time spinal endoscopic video segmentation problem. The result collection and processing for this study occurred in two phases. The Phase 1 included the execution of six candidate attention mechanisms across five different positional distributions for 50 epochs with three different random seeds, yielding a total of 66 runs. Phase 2 included 100 epochs with 10 random seeds, yielding 100 total trials to validate 10 mechanisms selected to assign attentional positions. The dataset used to conduct the study included 6,958 annotated frames of six anatomical structures. Overall, it took approximately 10 weeks of complete training run time, with an overwhelming majority of total run time coming from training time in Phase 2.

6.2 Main Findings

The central finding is that attention mechanism effectiveness is position-dependent rather than uniform. Triplet Attention provides statistically significantly improved accuracy for both L19 (high resolution layer) and L23 (medium resolution layer), but not for L27 (low resolution semantic layer). The optimal position-aware configuration within the configurations tested, Hybrid-L15CA, places Channel Attention at the structurally distinct FPN fusion layer L15, Triplet Attention at L19 and L23, and retains the MSCA baseline at L11 and L27. The absolute best performance position-aware configuration we tested, Hybrid-L15CA (which has Channel Attention at L15; Triplet Attention at L19 and L23; and maintains an MSCA baseline at L11 and L27), had a mAP50-95 score of 0.5913 (+2.82%) compared to uniform deployment of Triplet Attention (0.5911), with only three of five positions modified. It achieved 95.9% of the performance of full uniform CA (0.5920) with less variance in training (CV = 0.99%, compared with 1.13%). Statistically significantly improved performance on five of the six anatomical structures studied, with the greatest improvement in the clinically significant Herniation anatomical structure (+5.60%), and a 66.2% increase over the performance of the same structure with full CA (+3.37%). All mechanisms modified by an attention have higher inference speeds than the MSCA baseline, with Hybrid-L15CA achieving 191.6 FPS, an 18%

increase in speed and an increase in accuracy. Overall, given the results of this study, we believe the common practice of deploying uniform attention is an inadequate default and needs to be re-evaluated in the context of multi-scale architectures and when integrating attention mechanisms into them.

In addition to the overall primary result, there were three other important and supporting findings that warrant discussion. First, Phase 1 experimentation produced 50 epoch rankings that changed direction at the completion of full convergence for a number of different configurations (including L27 - Triplet); therefore, training for sufficient time and repeating the training across sufficiently many random seeds is essential for establishing reliable results when making sub-percentage comparisons between similar architectures. Secondly, the nearly perfect additive relationship between the improvements at L19 and L23 (101.7%) has shown that both positions yield complementary representations of non-redundant improvements across the segmented frames. Thirdly, although it was found that the tested alternative attention mechanisms failed at L27 and L11 but succeeded in other locations, this is empirically the theoretical argument that each type of attention mechanism has a preference based on the feature representations at each position.

6.3 Limitations & Future Work

The most significant limitation of the present study is that we only had access to data from a single institution where only one clinical dataset was identified therefore it is not known how well these mechanistic functions for Hybrid-L15CA will work with different styles of surgical videos, different targets & scales of datasets throughout the surgical community. Phase 2 of the study only tested four initial families of mechanisms, therefore there may be more suitable mechanisms for Hybrid-L15CA's design that were not included in this initial scope.

There are three major areas for future work. 1) Validating Hybrid-L15CA on laparoscopic, colonoscopic and robotic-assisted surgical video datasets to determine whether the positional preferences for Hybrid-L15CA obtain from spinal endoscopy can also be observed with surgical data from other surgical modalities. 2) Attention maps produced from each of the five attention positions may provide insight into the relationship between their accuracy conversion and the actual functions of the predicted Mechanisms. 3) The systematic exploration of hybrid neural architecture combinations for the purpose of finding co-optimized mechanisms and positional configurations can produce a better understanding of the general combinatorial nature of these types of neural networks than the two phased manual search

processes employed in this study and possibly establish further enhancements to both L11 and L27 that were not previously identified through the candidate Mechanisms tested in this investigation.

6.4 Reflection

The most impactful result in this project was not the best result; it was the null result from L27-Triplet. The same mechanism that allowed for consistent and statistically significant improvement at L19 and L23 produced no benefit at L27 due solely to positional differences. Until I observed how this positional aspect influenced results, I considered my choice of attention mechanisms to be largely a matter of preference, but the data made it necessary for me to view it as having a positional basis rather than as an after-the-fact explanation. At 20×20 resolution, a spatial attention mechanism using 32×32-pixel receptive fields cannot discern between close-lying tissue boundaries (e.g., the borders of muscle) because those boundaries do not appear at that level of resolution, and the failure to produce an expected benefit at L27 revealed what I now regard to be my clearest evidence for one of the major claims of the thesis.

The number of experimental runs (166) also caused me to alter my previous perspective regarding experimental rigor. The difference between the L27-CA values from Phase 1 and Phase 2 (+0.51% to almost 0.00%) compared to the L23-Triplet values illustrate that when trying to estimate the improvement (less than 1%), the estimate from 3 seeds is noise with a directional bias rather than a reliable measurement. By allowing additional time for computation to reach 10-seed estimates with a CV (Coefficient of Variation) value below 1%, I believe that this was the correct decision for the project. Interestingly, the differences in conclusions drawn from each method were greater than I anticipated, with the takeaway being that the methodological approach employed in stochastic systems with low signal-to-noise ratios determines whether or not the outcome will be considered real and applicable. I will certainly carry this knowledge with me into future projects rather than any particular technical finding.

When I look back on this project as a whole, the element of most value for me was the discipline to isolate the effects of the variables I manipulated. The C2f design used to develop the data enabled the following: entry and exit convolutions, channel dimensions and hyperparameters for each merge configuration were all held to the same values. Thus, it is possible to attribute all differences found in the final analyses to the chosen attention mechanism and its position only. This design decision was taken early in this project and at the time seemed like an unimportant design decision; however, it became evident during the course of the project

that this design choice was the reason that any of the ablation results could be interpretively determined. This is a broad lesson learned; in terms of experimental design, clean design is frequently of greater importance than the specific methods being tested.

References

- [1] C. A. Ikwuegbuenyi, S. Inzerillo, E. Wang, and I. Hussain, “Strategies for Optimizing Clinical Outcomes in Minimally Invasive Spine Surgery,” *Neurosurgery*, vol. 96, no. 3S, pp. S139–S147, Mar. 2025, doi: 10.1227/neu.0000000000003331.
- [2] M. Farshad *et al.*, “Classification of endoscopic spine procedures,” *North Am. Spine Soc. J. NASSJ*, vol. 22, p. 100603, Jun. 2025, doi: 10.1016/j.xnsj.2025.100603.
- [3] D. N. Kamtam *et al.*, “Deep learning approaches to surgical video segmentation and object detection: A scoping review,” *Comput. Biol. Med.*, vol. 194, p. 110482, Aug. 2025, doi: 10.1016/j.combiomed.2025.110482.
- [4] D. Palaniappan *et al.*, “YOLO in Healthcare: A Comprehensive Review of Detection Architectures, Domain Applications, and Future Innovations,” *IEEE Access*, vol. 13, pp. 145714–145735, 2025, doi: 10.1109/ACCESS.2025.3599358.
- [5] S. Y. Mohammed, “Architecture review: Two-stage and one-stage object detection,” *Frankl. Open*, vol. 12, p. 100322, Sep. 2025, doi: 10.1016/j.fraope.2025.100322.
- [6] D. Reis, J. Kupec, J. Hong, and A. Daoudi, “Real-Time Flying Object Detection with YOLOv8,” 2023, *arXiv*. doi: 10.48550/ARXIV.2305.09972.
- [7] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, “A Comprehensive Review of YOLO Architectures in Computer Vision: From YOLOv1 to YOLOv8 and YOLO-NAS,” *Mach. Learn. Knowl. Extr.*, vol. 5, no. 4, pp. 1680–1716, Nov. 2023, doi: 10.3390/make5040083.
- [8] J. Hu, L. Shen, and G. Sun, “Squeeze-and-Excitation Networks,” in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Salt Lake City, UT: IEEE, Jun. 2018, pp. 7132–7141. doi: 10.1109/CVPR.2018.00745.
- [9] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, “CBAM: Convolutional Block Attention Module,” in *Computer Vision – ECCV 2018*, vol. 11211, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., in Lecture Notes in Computer Science, vol. 11211. , Cham: Springer International Publishing, 2018, pp. 3–19. doi: 10.1007/978-3-030-01234-2_1.
- [10] D. Misra, T. Nalamada, A. U. Arasanipalai, and Q. Hou, “Rotate to Attend: Convolutional Triplet Attention Module,” in *2021 IEEE Winter Conference on Applications*

- of *Computer Vision (WACV)*, Waikoloa, HI, USA: IEEE, Jan. 2021, pp. 3138–3147. doi: 10.1109/WACV48630.2021.00318.
- [11] Y. Zhou, Y. You, X. Tan, and J. Tang, “AMAY-Net: adaptive multi-scale attention YOLO network for liver and gallbladder segmentation in laparoscopic cholecystectomy,” *PeerJ Comput. Sci.*, vol. 11, p. e2961, Aug. 2025, doi: 10.7717/peerj-cs.2961.
- [12] S. Frey *et al.*, “Optimizing intraoperative AI: evaluation of YOLOv8 for real-time recognition of robotic and laparoscopic instruments,” *J. Robot. Surg.*, vol. 19, no. 1, p. 131, Mar. 2025, doi: 10.1007/s11701-025-02284-7.
- [13] J. Yan *et al.*, “Enhanced object detection in pediatric bronchoscopy images using YOLO-based algorithms with CBAM attention mechanism,” *Heliyon*, vol. 10, no. 12, p. e32678, Jun. 2024, doi: 10.1016/j.heliyon.2024.e32678.
- [14] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature Pyramid Networks for Object Detection,” in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI: IEEE, Jul. 2017, pp. 936–944. doi: 10.1109/CVPR.2017.106.
- [15] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, “Path Aggregation Network for Instance Segmentation,” in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Salt Lake City, UT: IEEE, Jun. 2018, pp. 8759–8768. doi: 10.1109/CVPR.2018.00913.
- [16] H. Long *et al.*, “LC-YOLOmatch: a novel scene segmentation approach based on YOLO for laparoscopic cholecystectomy,” *Front. Artif. Intell.*, vol. 8, p. 1706021, Dec. 2025, doi: 10.3389/frai.2025.1706021.
- [17] A. Tashtoush, Y. Wang, M. T. Khasawneh, A. Hader, M. S. Shazeeb, and C. G. Lindsay, “Real-time object segmentation for laparoscopic cholecystectomy using YOLOv8,” *Neural Comput. Appl.*, vol. 37, no. 4, pp. 2697–2710, Feb. 2025, doi: 10.1007/s00521-024-10713-1.
- [18] Z. Xu *et al.*, “Spine endoscopic atlas: an open-source dataset for surgical instrument segmentation,” *Sci. Data*, vol. 12, no. 1, p. 1611, Oct. 2025, doi: 10.1038/s41597-025-05897-7.
- [19] H. Ma and G. Song, “A dual-branch medical image segmentation network based on a hybrid method,” *Comput. Eng. Des.*, vol. 47, no. 1, pp. 294–300, 2026, doi: 10.16208/j.issn1000-7024.2026.01.036.

- [20] R. Chen *et al.*, “MedFuseNet: fusing local and global deep feature representations with hybrid attention mechanisms for medical image segmentation,” *Sci. Rep.*, vol. 15, no. 1, p. 5093, Feb. 2025, doi: 10.1038/s41598-025-89096-9.
- [21] Z. Liu and J. Cheng, “CB-FPN: object detection feature pyramid network based on context information and bidirectional efficient fusion,” *Pattern Anal. Appl.*, vol. 26, no. 3, pp. 1441–1452, Aug. 2023, doi: 10.1007/s10044-023-01173-9.
- [22] G. Sun *et al.*, “DA-TransUNet: integrating spatial and channel dual attention with transformer U-net for medical image segmentation,” *Front. Bioeng. Biotechnol.*, vol. 12, p. 1398237, May 2024, doi: 10.3389/fbioe.2024.1398237.
- [23] X. Ning, Z. Gao, L. Huang, Z. Lü, and H. Jin, “MESAM-Net: A Multi-Branch Edge Convolution and Spatial Attention for 3D Edge Detection,” *J. Comput.-Aided Des. Comput. Graph.*, 2025, doi: 10.3724/SP.J.1089.2025-00240.
- [24] F. Li, H. Yan, and L. Shi, “Multi-scale coupled attention for visual object detection,” *Sci. Rep.*, vol. 14, no. 1, p. 11191, May 2024, doi: 10.1038/s41598-024-60897-8.
- [25] X. Wei, Z. Li, and Y. Wang, “SED-YOLO based multi-scale attention for small object detection in remote sensing,” *Sci. Rep.*, vol. 15, no. 1, p. 3125, Jan. 2025, doi: 10.1038/s41598-025-87199-x.
- [26] M. L. Ali and Z. Zhang, “The YOLO Framework: A Comprehensive Review of Evolution, Applications, and Benchmarks in Object Detection,” *Computers*, vol. 13, no. 12, p. 336, Dec. 2024, doi: 10.3390/computers13120336.
- [27] M. Saraei, M. Lalinia, and E.-J. Lee, “Deep Learning-Based Medical Object Detection: A Survey,” *IEEE Access*, vol. 13, pp. 53019–53038, 2025, doi: 10.1109/ACCESS.2025.3553087.
- [28] Z. Cai, K. Zhou, and Z. Liao, “A Systematic Review of YOLO-Based Object Detection in Medical Imaging: Advances, Challenges, and Future Directions,” *Comput. Mater. Contin.*, vol. 85, no. 2, pp. 2255–2303, 2025, doi: 10.32604/cmc.2025.067994.
- [29] M. Kang, C.-M. Ting, F. F. Ting, and R. C.-W. Phan, “ASF-YOLO: A novel YOLO model with attentional scale sequence fusion for cell instance segmentation,” *Image Vis. Comput.*, vol. 147, p. 105057, Jul. 2024, doi: 10.1016/j.imavis.2024.105057.

- [30] W. Han, X. Dong, G. Wang, Y. Ding, and A. Yang, "Application and improvement of YOLO11 for brain tumor detection in medical images," *Front. Oncol.*, vol. 15, p. 1643208, Aug. 2025, doi: 10.3389/fonc.2025.1643208.
- [31] A. Khan *et al.*, "A Recent Survey of Vision Transformers for Medical Image Segmentation," *IEEE Access*, vol. 13, pp. 191824–191849, 2025, doi: 10.1109/ACCESS.2025.3618215.
- [32] T. Goncalves, I. Rio-Torto, L. F. Teixeira, and J. S. Cardoso, "A Survey on Attention Mechanisms for Medical Applications: are we Moving Toward Better Algorithms?," *IEEE Access*, vol. 10, pp. 98909–98935, 2022, doi: 10.1109/ACCESS.2022.3206449.
- [33] Z. Lian *et al.*, "A Survey: Feature Fusion Method for Object Detection Field," in *Advanced Intelligent Computing Technology and Applications*, vol. 14864, D.-S. Huang, Z. Si, and Y. Pan, Eds., in *Lecture Notes in Computer Science*, vol. 14864. , Singapore: Springer Nature Singapore, 2024, pp. 84–95. doi: 10.1007/978-981-97-5588-2_8.
- [34] T. Kiobya, J. Zhou, and B. Maiseli, "A multi-scale semantically enriched feature pyramid network with enhanced focal loss for small-object detection," *Knowl.-Based Syst.*, vol. 310, p. 113003, Feb. 2025, doi: 10.1016/j.knosys.2025.113003.
- [35] Y. Feng, X. Zhu, X. Zhang, Y. Li, and H. Lu, "PAMSNet: A medical image segmentation network based on spatial pyramid and attention mechanism," *Biomed. Signal Process. Control*, vol. 94, p. 106285, Aug. 2024, doi: 10.1016/j.bspc.2024.106285.
- [36] Z. Al-Huda, M. A. Al-Antari, R. N. A. Algburi, O. Al-Maqtari, T. M. Rajeh, and G. A. Khan, "Triplet Attention-Enhanced UNet Architectures for Advanced Skin Lesion Segmentation," in *2024 8th International Artificial Intelligence and Data Processing Symposium (IDAP)*, Malatya, Turkiye: IEEE, Sep. 2024, pp. 1–8. doi: 10.1109/IDAP64064.2024.10710922.
- [37] Q. Hou, D. Zhou, and J. Feng, "Coordinate Attention for Efficient Mobile Network Design," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Nashville, TN, USA: IEEE, Jun. 2021, pp. 13708–13717. doi: 10.1109/CVPR46437.2021.01350.
- [38] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks," 2019, *arXiv*. doi: 10.48550/ARXIV.1910.03151.

- [39] O. Rainio, J. Teuho, and R. Klén, “Evaluation metrics and statistical tests for machine learning,” *Sci. Rep.*, vol. 14, no. 1, p. 6086, Mar. 2024, doi: 10.1038/s41598-024-56706-x.
- [40] J. Cohen, *Statistical power analysis for the behavioral sciences*, 2nd ed. Hillsdale, N.J: L. Erlbaum Associates, 1988.
- [41] C. Urrea, Y. Garcia-Garcia, and J. Kern, “Improving Surgical Scene Semantic Segmentation through a Deep Learning Architecture with Attention to Class Imbalance,” *Biomedicines*, vol. 12, no. 6, p. 1309, Jun. 2024, doi: 10.3390/biomedicines12061309.
- [42] G. Ayana, K. Dese, A. M. Abagaro, K. C. Jeong, S.-D. Yoon, and S. Choe, “Multistage transfer learning for medical images,” *Artif. Intell. Rev.*, vol. 57, no. 9, p. 232, Aug. 2024, doi: 10.1007/s10462-024-10855-7.
- [43] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, “RandAugment: Practical automated data augmentation with a reduced search space,” 2019, *arXiv*. doi: 10.48550/ARXIV.1909.13719.
- [44] A. S. Talaat, M. A. Elsabagh, M. T.-H. Alouane, and E. Hassan, “Boosting YOLO-based architecture for blood cell segmentation with ONNX acceleration and mosaic data augmentation,” *Egypt. Inform. J.*, vol. 34, p. 100956, Jun. 2026, doi: 10.1016/j.eij.2026.100956.
- [45] T. Yang, X. Lu, L. Yang, M. Yang, J. Chen, and H. Zhao, “Application of MRI image segmentation algorithm for brain tumors based on improved YOLO,” *Front. Neurosci.*, vol. 18, p. 1510175, Jan. 2025, doi: 10.3389/fnins.2024.1510175.

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